

Composition and date in the Pentateuchal sources: what a linguistic model recovers, and what archaizing would cost

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Abstract

Linguistic dating assumes that a late author cannot convincingly write like an early one, and that the language of a text carries its date rather than its subject. Both are tested here on 144,517 words of anchored Biblical Hebrew, partitioned into 281 passages of about 500 words. Genre, not chronology, proves to be the dominant axis of variation: 77% of the variance in the model's predictions lies between genres, and rank correlation with true date falls from +0.666 to +0.405 once genre is held fixed. A permutation null shuffling dates only within genre confirms that a chronological signal survives ($p = 0.017$), but the surviving signal is entirely lexical—refitting on morphosyntax alone gives +0.140—and includes the classical-to-late first-person pronoun shift, recovered unsupervised in the correct direction. Across 44 analytic specifications the Deuteronomic and Priestly sources fall after the exile in every one and the JE composite in 41, in the classical order $JE \rightarrow D \rightarrow P$, with central estimates in the Persian period and intervals too wide to quote any single date. Within the sources, D is measurably the most homogeneous; but the apparent separations between each source's conventional blocks are significant on the full feature set and dissolve under the genre control, the Deuteronomic one reversing. They are register effects, not dates. Archaizing was tested separately, with opposite outcomes in the two languages. Substituting the recognized Classical forms throughout six securely dated late Hebrew books moves their estimates by 9 years, which is 7% of this model's residual scale: a non-detection, not a measurement that archaizing is cheap. In Greek, where the archaizing is real and its practitioners known, the Second Sophistic Atticizers are placed 206 years too early—202 against texts of their own genre—by a model trained on non-archaizing Greek. Deliberate imitation can displace a linguistic date by centuries, and the poems dated here should be read with that possibility open.

Introduction

Scholars in biblical studies, classical philology, and Near Eastern linguistics routinely use internal linguistic evidence to constrain the dates of ancient texts. The transition from Classical Biblical Hebrew (CBH; also called Standard Biblical Hebrew) to Late Biblical Hebrew (LBH) is among the best-documented diachronic shifts in any ancient language [1–3], and comparable transitions are well attested in Ancient Greek [4] and Akkadian. The practice has an obvious appeal for the Pentateuch, whose sources carry no colophons and whose dates have been argued from literary and historical grounds for two centuries.

I set out to produce calibrated dates for those sources. The main finding of this article is that the linguistic evidence supports much less than the literature assumes, and that the largest obstacle is not the one the field has been arguing about.

That obstacle is genre. In a corpus assembled by date, genre rather than chronology turns out to be the dominant axis of variation: 77% of the variance in the model’s predictions lies between genres, and rank correlation with true date falls from +0.666 to +0.405 once genre is held fixed. A permutation null that shuffles dates only within genre—preserving the genre–date association the corpus happens to contain and destroying only the within-genre chronology—confirms that a chronological signal survives ($p = 0.017$). But it is roughly half the raw figure, and studies in this literature that report raw correlations without a genre control are overstating their evidence. The surviving signal is moreover entirely lexical: refitting on verb morphology, clause structure and agreement, with every lexeme removed, gives +0.140. Whatever grammatical change occurred over this period is not detectable at this sample size.

What the evidence does support is a placement claim. Across 44 defensible analytic specifications the Deuteronomistic and Priestly sources fall after the exile without exception, and the JE composite in 41 of them, in the classical sequence $JE \rightarrow D \rightarrow P$, with central estimates in the Persian period. The intervals are roughly 204 yr wide at 68% on either side, and no individual date in this study should be quoted to a century. I report the full specification curve rather than a preferred configuration, because no configuration here has a strong claim to being the right one.

Within the sources the picture is mixed, and the negative half of it is worth as much as the positive. The Deuteronomistic source is measurably the most internally homogeneous of the three, and that holds however hard the genre-diagnostic features are screened out. But the separations between each source’s conventional blocks—the Deuteronomistic law code from its narrative frame, the Holiness Code from the Priestly material of Leviticus 1–16, the JE material of Genesis from that of Exodus—behave differently. On the full feature set all three are significant and all three point the way the critical literature would predict. Under the genre control they vanish, and the Deuteronomistic separation reverses and becomes significant in the opposite direction. The law code is legal prose and its frame is hortatory narrative; the gap between them was carried by the vocabulary that distinguishes those registers. I report this at length because the unscreened numbers are exactly what a study would publish if it stopped one step early, and because it shows what the genre control is for.

Beneath the whole enterprise sits an assumption that is stated often and tested never. Linguistic dating presupposes that a late author cannot convincingly write like an early one. Were deliberate archaizing easy, so that a Persian-period scribe could adopt Classical vocabulary and thereby acquire a Classical date, every linguistic argument for the antiquity of a biblical text would be worthless, and the objection would apply to traditional and computational methods alike. The dialectal school has pressed this point for two decades [5, 6], while the mainstream reply has been that sustained archaizing is difficult. Both positions are assertions. Neither side has put a number on the matter.

Archaizing is therefore tested here twice, and the two tests do not come out the same way. I can perform archaizing myself, so the ground truth is known exactly: securely dated *late* books have their LBH forms replaced by CBH counterparts at controlled rates, every feature is re-extracted from the altered text, and the result is dated by a model that never saw it. The displacement this produces is far below the model’s own noise, so what the Hebrew experiment establishes is that archaizing of that kind cannot be detected here—not that it would be cheap for a scribe. Ancient Greek then supplies what Hebrew cannot, namely texts whose authors really did archaize, deliberately and expertly, and whose dates of composition are securely known. The Second Sophistic Atticizers wrote imitation Classical Attic four to seven centuries after the fact, taught by a formal curriculum, well enough to be read as models of style for a millennium. This

prices the *morphosyntactic* mechanism, and the price is an order of magnitude higher. The Greek measurement is put through the same genre control as the Hebrew one, and survives it: Atticizing texts are dated 202 yr earlier than non-archaizing texts of the same genre.

That contrast is what this study offers against the standing objection, and it is narrower than a price list. Where archaizing is real and documented it displaces a linguistic date by centuries, which settles that the objection has teeth. Where it is simulated by substituting diagnostic vocabulary, this model cannot see it at all, which settles only that this model is the wrong instrument for that question. Whether a given text could have been archaized therefore turns on what its author was trained to do—a historical question about scribal education, not a statistical one—and on an instrument more sensitive than this one.

The Background section reviews the CBH/LBH distinction, the dialectal challenge, and the problem of circularity. The Data section describes the anchored corpus, the passage-level design, and feature extraction. The Model section presents the weighted ridge estimator, the variance-matching step, and the jackknife+intervals. The Validation section documents a failure mode that inflates reported accuracy in this literature, establishes the genre confound and the lexical character of the signal, and reports leave-one-book-out performance against two permutation nulls. The two Results sections report the internal structure of the sources and their placement relative to the exile, and the section following them reports both archaizing measurements.

Background: Diachronic Biblical Hebrew

The CBH/LBH Distinction

Classical Biblical Hebrew (CBH) is the language of the pre-exilic prophetic corpus and early narrative material (8th–7th centuries BCE): Amos, Hosea, early Isaiah, and the narrative strands of Samuel–Kings. Late Biblical Hebrew (LBH) characterizes the post-exilic prose of Ezra, Nehemiah, Chronicles, Esther, and Daniel (5th–2nd centuries BCE).

The transition is marked by well-studied lexical, morphological, and syntactic changes [2, 3, 7, 8]. At the lexical level, the first-person singular pronoun shifts from *ankī* to *anī*; the relative pronoun *ašer* gives way to the enclitic *še-*; the negator *lō'* is supplemented by and partly replaced by *'ēn*. At the morphological level, the wayyiqtol declines in frequency while the *hāyāh* + participle periphrastic increases. At the syntactic level, infinitive-absolute constructions rise, and sentence structure shifts toward analytic patterns associated with Aramaic influence [5, 6].

The consensus view [1, 9–11] holds that these features shift monotonically along the temporal axis, constituting a reliable if noisy diachronic signal. The present study adopts this view as a working assumption and then tests the principal alternative to it directly.

Diachronic or Dialectal

Young, Rezetko, and Ehrensverd [5] argue that the CBH/LBH contrast may reflect geographic dialect, scribal tradition, or literary register rather than chronology. The Qumran sectarian texts (~150 BCE) deliberately employ CBH vocabulary and morphology while simultaneously using Aramaic loan-words — a pattern they interpret as literary register rather than temporal reality. Hurvitz and the diachronic mainstream [7] counter that the dialectal alternatives require additional auxiliary hypotheses, and that the Qumran archaism case presupposes the original CBH features had already receded, confirming rather than undermining the diachronic interpretation.

This debate has been conducted almost entirely through argument. Its empirical core, however, is a quantitative question: how much apparent antiquity does deliberate archaizing actually purchase? That question is answerable. A model that assigns dates can be fed texts whose archaizing status is known, either because I performed the archaizing myself or because the author's imitative program is a matter of literary-historical record. Both experiments are reported below. Neither resolves the dispute, but each supplies a measurement that either side can now argue from, which is more than the debate has had so far.

The Problem of Circularity

Dating methods that use features identified by comparing texts of assumed date risk circularity: the assumed dates shape the feature set, and the feature set is then used to infer unknown dates. Three safeguards are adopted.

A distinction is necessary before the safeguards can be stated, because the objection applies with very different force to different kinds of evidence. Dating a text by its resemblance to other Hebrew of supposed date is circular for the present purpose, since it invokes the very cline the model is meant to estimate. Dating a text by a datable referent is not. Esther’s Achaemenid court and administration, the fall of Nineveh behind Nahum, and the Persian administrative vocabulary of Ecclesiastes are all anchored to events and language contacts documented outside the biblical tradition. That Esther cannot be pre-exilic follows from Persian history, not from Hebrew style, and the inference would stand were the language of Esther entirely unremarkable. The same logic underwrites Ezekiel’s regnal dates. They are evidence because the kings and years can be checked against Babylonian records; had he invented them, they would be worth nothing.

The circularity that has damaged linguistic dating runs in the other direction, and it runs through the Torah. The pre-exilic standard against which lateness is measured was itself constructed in part from Torah texts assumed to be pre-exilic, and those texts were then dated by their conformity to it [5,6]. Nothing in the present design enters that loop. The model learns the feature-to-date mapping from the anchored corpus alone, no Torah text is used in training, and every Torah estimate is therefore a genuine out-of-sample prediction.

With that distinction in place, three safeguards apply.

First, every training date rests on a synchronism or on a datable historical referent. Twelve of the 25 units carry an explicit synchronism, a named ruler or regnal year checkable against Assyrian, Babylonian, or Persian records; twelve more are fixed by a datable event, institution, or language contact. One unit, Jonah, is dated principally on typological grounds, and its removal is reported in Table 2.

Second, no feature is selected using date information. The feature set is fixed in advance by frequency rank in the corpus as a whole (Data section), not by correlation with date. Feature-by-date screening was evaluated and rejected: at this dimensionality the false discovery proportion among screened features is high enough that screening manufactures apparent signal (S2 Table).

Third, all reported performance comes from leave-one-*book*-out cross-validation in which standardization, weighting, regularization-parameter selection and interval calibration are performed inside each fold, and significance is established by a permutation null that re-runs the entire pipeline on shuffled dates.

Why an anchored corpus. The Hebrew prophetic books offer a form of historical anchoring available nowhere else in the Hebrew Bible: their superscriptions and internal chronological notices can be cross-checked against records entirely external to the biblical tradition. Ezekiel explicitly dates thirteen of his visions to regnal years of the Babylonian exile — “the fifth year of King Jehoiachin’s captivity” (Ezek 1:2), “the sixth year” (8:1), and so forth — which correlate with Babylonian administrative texts independently dating Jehoiachin’s deportation to 597 BCE. Other eighth- and seventh-century prophets (Amos, Hosea, Isaiah, Micah, Zephaniah, Nahum) are anchored by superscriptions naming Judahite and Israelite kings whose reigns are documented in the Assyrian and Neo-Babylonian annals. No comparable external anchoring exists for the Torah or the poetic books: their dates derive entirely from literary-critical argument, which is precisely the circularity this study seeks to avoid.

Asymmetric error and conservative upper bounds on antiquity. If the accepted dates for the anchored training texts were wrong, the error would be asymmetric: these texts can be *no older* than the historical events they reference, but they could in principle be *younger*. A training corpus misdated in the direction of being too early would calibrate the feature-to-date mapping at too early a point on the temporal axis, so unknown texts dated against that calibration would also be estimated as too early. Any systematic error in the training dates therefore propagates in a single direction: all model-derived dates shift toward greater recency. The dates reported here should be read as *conservative upper bounds on antiquity*. A text dated to 400 BCE may be more recent; the linguistic evidence provides no support for a date earlier than the estimate.

Prior Computational Work: Attribution versus Dating

Computational approaches to textual attribution and stylometric dating have a substantial history, from Mosteller and Wallace’s Bayesian analysis of the Federalist Papers [12] through the stylometric tradition surveyed by Holmes [13] and the function-word methods of Burrows [14]. Among work applied specifically to the Hebrew Bible, Faigenbaum-Golovin et al. [15] provide the most methodologically rigorous prior computational treatment. Their Higher Criticism discrepancy method tests whether the distribution of lemma n-gram frequencies in a target text deviates from the null that it was produced by the same process as a reference corpus. Applied to 50 ground-truth chapters spanning D, the Deuteronomistic History, and P, it achieves 84% attribution accuracy in leave-one-out validation and yields interpretable discriminating lemma lists.

The present framework answers a different question, and three differences follow from that.

Continuous dates versus categorical assignment. The Higher Criticism method assigns each text to the most likely of three corpora. However, D and the Deuteronomistic History are both dated to the late monarchic or exilic period, so two texts composed at 650 and 590 BCE receive identical labels, and no intra-school temporal resolution is possible. The model used here returns a date with a calibrated interval, recovering whatever temporal resolution the data support and quantifying the rest as uncertainty.

Period features versus topic features. Faigenbaum-Golovin et al. operate on lemma and lemma n-gram frequencies, a representation in which content words — *gold* in Tabernacle chapters, *king* in regnal narratives — dominate. Content-word frequency is sensitive to topic and genre as much as to date. The feature set here is deliberately dominated by high-frequency grammatical rates (verbal aspect and stem distributions, part-of-speech transition rates, clause-relation profiles) that vary with period rather than subject matter. This is what permits comparison of poetry against narrative, which a bag-of-words representation cannot do.

An explicit accounting of archaizing. Attribution methods offer no mechanism for telling a genuinely ancient text from one composed in a deliberately archaic register. A Persian-period author who archaizes to D norms will be assigned to D regardless of when he wrote. Rather than claim immunity from the problem, the sensitivity is tested directly below. In Greek, where the archaizing is real and its authors are known, it is measured and it is large. In Hebrew the test returns a non-detection: the displacement produced by substituting the recognized diagnostic vocabulary is well below this model’s noise floor, so the honest report is that archaizing of that kind cannot be detected here rather than that it is cheap.

These are differences of purpose rather than deficiencies in either approach. The ideal treatment of a disputed text would apply both, as attribution output can set register priors while date estimates constrain the plausible settings for an attribution decision.

Data: Corpus and Feature Extraction

The BHSA Database

All Hebrew analysis uses the Biblia Hebraica Stuttgartensia Amstelodamensis (BHSA) database [16], accessed through Text-Fabric [17]. Extraction is restricted to words that BHSA marks as Hebrew, which matters more than it sounds: Ezra 4:8–6:18 and 7:12–26 are Aramaic and account for 28.9% of that book, and Jer 10:11 adds a further 19 words. Daniel’s Aramaic section, 2:4b–7:28, is excluded by the chapter range assigned to that unit. BHSA supplies a complete morphological and syntactic annotation of the Masoretic text: for every word, a lexeme identifier, part of speech, verbal tense/aspect and stem, person, number, gender, pronominal suffix and state; and for every phrase and clause, a type, a syntactic function, and a relation to its governing clause. These annotations are the product of decades of work by the Eep Talstra Centre for Bible and Computer and are independent of any dating hypothesis.

The Anchored Corpus

The training corpus comprises 25 units spanning 760–167 BCE, listed in Table 1. Books with disputed internal composition histories are split into the conventional scholarly divisions (Isaiah into three, Zechariah

into two), and the oracular core of Jeremiah is separated from its Deuteronomistic prose frame, which is treated as an undated target rather than an anchor.

The units differ in the kind of evidence that fixes their dates, and the difference is marked in the table along the lines set out above. Twelve carry an explicit synchronism and twelve a datable historical referent, neither of which depends on the feature cline under analysis. Jonah alone is dated principally by typology, following the critical commentary [18], with some recent work pressing it later still [19]; even there a terminus post quem is available, since Nineveh is described as a city that already lay in ruins (Jonah 3:3).

Table 2 reports how much any of this matters. A leave-one-unit-out jackknife, re-running the entire pipeline 25 times, gives a rank correlation between +0.48 and +0.76 with a median of +0.65, and pairwise accuracy between 66.4% and 78.6%. That spread is the leverage a single unit carries at this corpus size, and it falls almost entirely on the late anchors that fix the recent end of the scale: removing Nehemiah, Esther, Daniel, Ezra, or Jonah each costs between 0.14 and 0.17 of correlation, while removing any pre-exilic unit costs at most 0.05. The full-corpus figure of +0.66 should accordingly be read against that range rather than as a point estimate.

Removing the less securely anchored units changes little. Dropping all six that lack an explicit synchronism leaves ρ at +0.59 with a mean absolute error of 111 yr against a 127 yr baseline, better in both respects than the full corpus. Dropping Jonah and Ecclesiastes together costs 0.19 of correlation, which is what dropping any two late anchors costs and is not peculiar to them.

What matters more than whether the anchors might be wrong is which way being wrong would push the results, and that is not symmetric. The model learns its mapping from language to date from the anchors, so an anchor whose true date is *later* than the date assigned to it teaches the model to associate that book's linguistic profile with too early a date, and the error is carried into every target whose language resembles it. Anchor error propagates to the targets with the same sign.

Displacing the whole chronology 50 yr toward the present and re-running the estimator moves the three sources by -50, -50 and -50 yr: pass-through is 100% and one for one. Confining the displacement to the six anchors that lack an external synchronism—the case where the securely fixed dates are right and only the judgment calls are wrong—moves them by at most 24 yr, so the units whose dating is most open to argument carry correspondingly little of the result. Displacing only the post-exilic anchors moves the sources -54, -54 and -64 yr, slightly more than the displacement applied, since it is the late anchors that fix where late-looking targets land.

The direction of any likely error therefore matters, and it is worth stating which way the field has been moving. A century of critical work has revised biblical dates later, not earlier, and the recent archaeological argument [20] runs the same way. If the anchors used here are systematically wrong, they are more probably assigned too early than too late. Since the conclusion drawn below is that the Pentateuchal sources are *late*, an anchor chronology erring early makes that conclusion conservative: correcting it would move the sources later still. The error that would threaten the conclusion is the opposite one, and it is the less likely of the two. This is an argument about direction, not a correction, and no adjustment is applied anywhere in what follows.

Why Passages Rather Than Books

The most consequential design choice in this article is that features are extracted from passages of about 500 words rather than from whole books.

At the level of whole books the corpus offers only 25 observations. Any feature set large enough to describe Hebrew morphosyntax then has more dimensions than there are data points. The standard responses to this — screening features by their correlation with date, or reducing dimension using the date labels — both consult the labels before cross-validation and thereby inflate apparent performance. Worse, a whole-book rate is a single number summarizing tens of thousands of words, so a book whose passages are internally consistent cannot be told apart from one that is a composite.

Partitioning at ~500 words yields 281 passages from the same 144,517 words. This is not a free multiplication of the sample size. Passages from one book are correlated, so the appropriate measure is the effective sample size $n_{\text{eff}} = n / (1 + (\bar{m} - 1)\rho_I)$, where ρ_I is the intraclass correlation and $\bar{m}$ the mean number of passages per book. Across passage sizes from 300 to 1000 words the median intraclass correlation runs 0.09–0.20, and the effective sample size is 3 to 7 times the book-level figure (S2 Table). That is the real gain,

Table 1. The training corpus. Dates are BCE; “Pass.” is the number of ~500-word passages. Type S dates rest on an explicit synchronism, a named ruler or regnal year checkable against Assyrian, Babylonian or Persian records. Type H dates rest on a datable event, institution, or language contact documented outside the biblical tradition. Neither depends on the feature cline this model estimates. Type T marks the one unit dated principally by resemblance to Hebrew of supposed date; Table 2 reports the effect of removing it.

Unit	Date	Words	Pass.	Type	Basis of the date
Amos	760	2,780	5	S	Superscription: Uzziah of Judah / Jeroboam II of Israel
Hosea	745	3,146	6	S	Superscription: Uzziah to Hezekiah / Jeroboam II
Micah	725	1,895	4	S	Superscription: Jotham, Ahaz, Hezekiah
Isaiah_1	720	13,504	26	S	Superscription: Uzziah to Hezekiah; Assyrian campaigns
Nahum	650	746	1	H	Fall of Nineveh, 612 BCE
Zephaniah	630	1,037	2	S	Superscription: Josiah
Habakkuk	605	897	2	H	Neo-Babylonian ascendancy
Jer_oracle	605	14,520	28	S	Superscription: Josiah to Zedekiah; 597 and 586 BCE
Obadiah	585	392	1	H	Edomite conduct at the fall of Jerusalem, 586 BCE
Ezekiel	580	26,182	51	S	Thirteen regnal dates from Jehoiachin’s deportation, 597 BCE
Lamentations	570	1,945	4	H	Destruction of Jerusalem, 586 BCE
Isaiah_2	545	5,738	11	S	Cyrus named; fall of Babylon, 539 BCE
Haggai	520	877	2	S	Regnal dates, second year of Darius I, 520 BCE
Zechariah_1	518	2,484	5	S	Regnal dates, second to fourth year of Darius I
Malachi	460	1,187	2	H	Persian governor (<i>peha</i>); second-temple cult
Joel	400	1,318	3	H	Functioning second temple; no monarchy
Jonah	400	985	2	T	Nineveh already fallen (3:3); post-exilic novella
Isaiah_3	400	3,689	7	H	Restored temple cult; Persian-period institutions
Ezra	380	3,745	7	S	Persian regnal dates, Cyrus to Artaxerxes
Nehemiah	380	7,842	15	S	Twentieth year of Artaxerxes I, 445 BCE
Zechariah_2	350	1,987	4	H	Greek (<i>Yawan</i>) as a military power
Chronicles	330	35,330	69	H	Genealogies to the Persian period; Cyrus decree
Esther	300	4,621	9	H	Achaemenid court and administration; Xerxes
Ecclesiastes	250	4,233	8	H	Persian loanwords (<i>pardes</i> , <i>pitgam</i>)
Daniel	167	3,437	7	H	Antiochene persecution, 167 BCE

and it is what makes a feature set of this width estimable at all.

Passage size trades two effects against each other. Smaller passages give more observations but sparser features, since a lexical item occurring once per thousand words will be absent from most 200-word passages, and its rate is then not a measurement but a coin flip. Larger passages give denser features but fewer observations. I therefore crossed passage sizes of 200, 300, 400, 500, 700, and 1000 words with three feature sets and with calibration on and off, 36 configurations in all, selecting on out-of-sample rank correlation under leave-one-book-out (S3 Table). The selected configuration uses ~500-word passages, though performance is flat between 400 and 1000, so the choice is not a delicate one.

Passages respect verse boundaries: words are accumulated verse by verse until the target is reached. A trailing remainder shorter than half the target is merged into the preceding passage rather than kept as a

Table 2. Robustness of the headline statistics. MAE and baseline are in years, the baseline being the constant predictor on the same subset. The jackknife drops each anchored unit in turn and re-runs the entire pipeline on the remaining 24; its spread is the leverage any single unit carries at this corpus size, and the full-corpus ρ should be read against it rather than as a point estimate. The lower block removes the units whose dates are least independent of the language: Jonah, the one unit dated principally on typological grounds, together with Ecclesiastes, and then all six units lacking an explicit synchronism.

Fit	Units	MAE	Baseline	ρ	Pairwise
Full corpus	25	118	137	+0.66	72.9%
<i>Leave-one-unit-out jackknife, all 25 fits</i>					
median	24	—	—	+0.65	72.3%
range	24	—	—	+0.48 to +0.76	66.4–78.6%
<i>Removing the less securely anchored units</i>					
without Jonah and Ecclesiastes	23	121	130	+0.51	66.4%
without all 6 non-synchronism units	19	111	127	+0.59	69.2%

short, high-variance observation. The resulting passages are close to the nominal size, from 312 to 757 words with a median of 508 and 96% falling between 400 and 700. Two units, Obadiah and Nahum, are short enough to yield a single passage and are therefore book-level observations; their passage-level dispersion in Table 4 is zero for that reason rather than because they are homogeneous.

Cutting at verse boundaries also makes the phrase- and clause-level features well defined. Those features are collected over every phrase and clause overlapping a passage, so an object straddling a boundary would be counted in both neighbors and its type credited to each. In BHSa this scarcely happens: across the whole database only 0.01% of phrases and 0.06% of clauses span a verse boundary, and in the 281 passages of the training corpus the number of partially contained phrases and clauses is 0 and 0 respectively. Sentences are the exception, spanning verses in 1.4% of cases overall and leaving 16 partially contained passages here (0.1%), which biases the mean sentence length by -0.2% .

Feature Extraction

The feature set is fixed by frequency rank computed over the whole BHSa corpus, without reference to date. 596 features are extracted, of which 18 are structurally constant on this corpus and are dropped inside each fold, leaving 578 in six families:

1. **Lexical rates** (250 features): occurrences per 1000 words of each of the 250 most frequent lexemes in the corpus. Since selection goes by corpus frequency, these are dominated by function words — articles, prepositions, conjunctions, pronouns — rather than by the rare diagnostic vocabulary of the CBH/LBH literature. This is deliberate, as the sparsity constraint above rules out the rare items, and the consequences of that are taken up below.
2. **Part-of-speech transition rates** (207 features): rates of each ordered pair of parts of speech across adjacent words, capturing local syntactic profile without reference to any particular construction.
3. **Verb morphology** (57 features): distributions over tense/aspect (qatal, yiqtol, wayyiqtol, participle, infinitive, imperative), stem (qal, nifal, piel, hifil, hitpaal, pual, hofal), and their interaction.
4. **Phrase and clause structure** (42 features): phrase type and syntactic function rates, and clause-relation rates from the BHSa clause hierarchy.
5. **Agreement and suffix morphology** (19 features): person, number, gender and pronominal-suffix rates.
6. **Structural** (3 features): mean phrase and clause length, clauses per passage, and type-token ratio.

Two extensions were built and evaluated and neither is used. A character and part-of-speech n-gram extension adding 689 further features does not improve out-of-sample performance at the selected passage

size (S3 Table). A clause-type family — rates and transition rates over the BHSA clause types that encode the Hebrew verbal system, including the wayyiqtol narrative clause (Way0) and the weqatal clause (WQt0) — was added after an audit found that an earlier implementation had counted clause types while indexing them by clause relation, whose value sets are disjoint, so that the family had been silently absent. Implemented correctly, its members are among the most strongly date-correlated single features in the corpus: 4 of the 14 clause types correlate with true date above $|\rho| = 0.5$ at book level, led by fronted yiqtol at 0.62 and infinitive construct at 0.55. And yet adding all 50 of them leaves out-of-sample performance unchanged ($\rho + 0.665$ to $+0.660$, mean absolute error 118 to 120 yr). Their information is already carried by the verb-morphology and part-of-speech transition features. The reported model is therefore the one without them, and the exercise is reported because it indicates the feature set is saturated rather than arbitrary.

Features whose standard deviation is zero on the training fold, or which are missing in more than 20% of passages, are dropped inside the fold. Remaining missing values are imputed at the training-fold median.

Model

Weighted Ridge Regression

Let $\mathbf{X}$ be the $n \times p$ matrix of passage features and $\mathbf{y}$ the vector of book dates, each passage inheriting its book's date. Features are centered and scaled on the training fold. The estimator is ridge regression,

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^n w_i (y_i - \mathbf{x}_i^{\top} \beta)^2 + \lambda \|\beta\|^2, \quad (1)$$

solved in dual form, $\hat{\beta} = \mathbf{A}^{\top}(\mathbf{A}\mathbf{A}^{\top} + \lambda\mathbf{I})^{-1}\tilde{\mathbf{y}}$, where $\mathbf{A}$ is the weighted standardized design. Because $p > n$ on the Hebrew side, the dual form is the cheaper route, and one eigendecomposition of $\mathbf{A}\mathbf{A}^{\top}$ serves the entire λ sweep. λ is selected by an inner leave-one-book-out loop nested inside each outer fold, never using the held-out book.

Inverse-Density Weighting

The anchored corpus is not uniform in time. Post-exilic material outnumbers pre-exilic material, and an unweighted fit is pulled toward the dense region: in the unweighted model no book receives a confidently pre-exilic estimate, which is a property of the sampling rather than of Hebrew. Weights are therefore taken inversely proportional to a kernel estimate of the date density,

$$w_i \propto \left[\sum_j \exp\left(-\frac{1}{2} \left(\frac{d_j - d_i}{h}\right)^2\right) \right]^{-\alpha}, \quad (2)$$

with bandwidth $h = 90$ yr and exponent $\alpha = 1$, normalized to mean one. Weighted ridge is implemented as ordinary ridge on rows scaled by $\sqrt{w_i}$.

Variance Matching and Its Cost

A regularized regression is a shrinkage estimator, so its predictions have systematically smaller variance than the quantity being predicted. Under plain ridge, the predicted dates for the anchored corpus span roughly a third of the true range, and no text can then be assigned a confidently early or confidently late date. This is an artifact of shrinkage rather than a finding about Hebrew. The correction is to match the first two moments of the out-of-sample predictions to the true dates,

$$\hat{d} = \bar{d} + S(\hat{d}_{\text{raw}} - \bar{d}_{\text{raw}}), \quad S = \frac{\text{sd}(d)}{\text{sd}(\hat{d}_{\text{raw}})}, \quad (3)$$

with $\bar{d}$, $\text{sd}(d)$, $\bar{d}_{\text{raw}}$ and S all computed from leave-one-book-out predictions on the training corpus only. Here $S = 1.79$. The step is the inverse-calibration correction familiar from the classical-versus-inverse regression

literature [21]: when the object is to recover the predictor rather than to minimize squared error in the response, matching the spread of the estimates to the spread of the quantity being estimated is the appropriate adjustment.

This is a deliberate trade, and both sides of it ought to be stated. Rescaling *increases* the mean absolute error, since inflating the spread of the predictions amplifies their errors along with their signal: the uncalibrated model gives 94 yr against 118 yr for the rescaled one. What the trade buys is a model that is two-sided. Uncalibrated predictions span only 294 yr against 593 yr of real dates and place just 4 of the 8 genuinely pre-exilic books before the exile. After rescaling, the predictions span 526 yr and 6 of 8 are placed correctly.

Because the substantive question in the Pentateuchal debate is which side of the exile a text falls on, a model that can hardly ever say “early” is useless for that purpose whatever its mean absolute error. Its inability to say “early” is a property of the estimator, not evidence that nothing is early. Readers whose interest is point accuracy rather than calibration should use the uncalibrated variant, whose figures are in S3 Table.

Computational Reproducibility

The analysis is a single pipeline of 29 stages, run by one script, in dependency order: corpus extraction from BHSA, feature construction, validation, the permutation nulls, the undated predictions, and every robustness analysis reported below. No number in this article is now produced by hand or at an interactive prompt; that this was not always true is the subject of the third paragraph.

Two properties are enforced, not merely claimed. First, every quantity in the text is a macro read from a result file at build time; a missing result file fails the build instead of leaving a stale figure in place, and a mechanical check refuses to compile the manuscript if any number appears in the prose that did not come from a result file. Second, the pipeline records a cryptographic digest of each of its 38 outputs and compares them on the next run, so that a change anywhere becomes visible at once, not at the next reading.

Running the pipeline twice from the corpus reproduces all 38 outputs byte for byte. Achieving that required fixing three defects that a casual re-run would not have exposed: a feature matrix that no longer matched the extractor that had supposedly produced it, three result files that had been generated at an interactive prompt and never committed, and a permutation checkpoint that survived a corpus rebuild and was silently reused against a model it no longer described. Each is the kind of error that changes published numbers without changing anything a reader can see. Long-running stages now record a fingerprint of their inputs beside their checkpoint and refuse to resume across a mismatch.

The verification log is included in the repository. Readers should note that reproducing the permutation nulls takes several hours; every other stage completes in about fifteen minutes.

Jackknife+ Prediction Intervals

Split conformal prediction assumes that every calibration residual comes from a single fitted model, on data exchangeable with the target [22,23]. That assumption does not hold here. Each residual in this design comes from a *different* model, fitted without its own book, which makes the procedure a jackknife rather than split conformal, and naive jackknife intervals carry no finite-sample guarantee. Intervals are therefore built by the jackknife+ of Barber et al. [24]. Writing μ_{-i} for the model fitted without book i and R_i for that book’s leave-one-out residual, the interval for a target x is

$$\left[q_{\alpha}^{-}\{\mu_{-i}(x) - R_i\}, q_{1-\alpha}^{+}\{\mu_{-i}(x) + R_i\} \right], \quad (4)$$

which is distribution-free and has finite-sample coverage of at least $1 - 2\alpha$. The price of the guarantee is that a nominal 90% interval is warranted only to 80% and a nominal 68% interval only to 36%; empirical coverage on the anchored corpus is 72% and 96% respectively, so in practice the intervals are conservative rather than optimistic.

The variance-matching constants are recomputed inside each fold, so that μ_{-i} and R_i are free of the book being predicted. Estimating them once on all 25 leave-one-out predictions, as an earlier version did,

understates the mean absolute error by about ten years and narrows the 68% half-width from 158 to 137 yr. Rank correlation and pairwise ordering are unaffected either way, since the rescaling is a positive linear map. For a target unit, $P(\text{post-exilic})$ is the fraction of the jackknife+ ensemble $\{\mu_{-i}(x) \pm R_i\}$ falling after 586 BCE.

Validation

A Failure Mode That Inflates Reported Accuracy

Reported accuracies for linguistic dating models of this class vary by two orders of magnitude. Much of that variation is not a difference in the underlying signal but an artifact of how the models are validated, and the artifact is worth naming because it is easy to commit and hard to see.

In a Bayesian dating model, the posterior over a held-out text’s date combines a likelihood from the linguistic features with a prior. Should the prior for the held-out text be centered on that text’s own scholarly date, which is the natural thing to do when the prior is meant to encode what scholars think, then the model is being asked to recover a number it has already been handed. The resulting holdout error measures the width of the prior rather than the information in the language.

The effect is large. Under a prior-centered design on this corpus, the median share of posterior precision contributed by the linguistic evidence is 11.7%, and mean holdout error is 44 yr. Under an agnostic prior with an otherwise identical pipeline, the same model achieves 156 yr. For the 8 units whose scholarly dates carry the tightest uncertainty ($\sigma \leq 20$ yr), the linguistic evidence supplies 4.5% of posterior precision and the leaky design reports 6.6 yr against an honest 189 yr (Table 3). For a model of this kind, a reported accuracy of a decade or two is a signature of the design rather than an achievement.

Table 3. Prior leakage inflates reported holdout accuracy. “Prior σ ” is the standard deviation of the scholarly prior in years; “data share” is the fraction of posterior precision contributed by the linguistic features; the two error columns are mean absolute errors in years under a prior centered on the unit’s own scholarly date and under an agnostic prior, with the pipeline otherwise identical.

Unit	Prior σ	Data share	Leaky err.	Honest err.
Haggai	10	2.3%	6	334
Daniel	10	1.7%	3	233
Zechariah.1	10	1.5%	4	236
Amos	20	5.8%	8	144
Habakkuk	20	5.4%	3	51
Isaiah.2	20	8.0%	15	267
Ezekiel	20	5.9%	10	158
Zephaniah	20	5.6%	4	90
Malachi	25	8.9%	18	192
Micah	25	9.7%	3	39
Nahum	25	8.6%	4	48
Hosea	25	9.2%	1	5
Isaiah.1	30	12.0%	16	140
Jer_oracle	30	11.7%	11	105
Chronicles	40	21.8%	20	116
Lamentations	50	35.8%	64	238
Jonah	60	34.1%	18	50
Ezra	60	33.8%	26	80
Esther	60	64.0%	80	200
Nehemiah	60	34.5%	36	116
Obadiah	80	49.2%	29	57
Isaiah.3	100	63.3%	186	296
Ecclesiastes	120	70.7%	150	212
Joel	150	78.3%	200	256
Zechariah.2	150	78.5%	190	242
Median, all	30	11.7%	16	144
Mean, $\sigma \leq 20$ (8 units)	16	4.5%	6.6	189.1

All results below use an agnostic prior; the reported model is not Bayesian and has no prior to leak.

The Leave-One-Book-Out Protocol

Cross-validation folds are whole books, never passages. Passages from one book are strongly correlated, so splitting them across a fold boundary would let the model see the held-out book’s own material. Within each fold, and using only the training books, features with zero variance or excessive missingness are dropped, medians for imputation are computed, standardization constants are computed, inverse-density weights are computed, λ is selected by an inner leave-one-book-out loop, and the variance-matching constants and jackknife+ residuals are derived. The held-out book’s date is used only to score the result.

A book’s estimate is the median of its passage estimates. The dispersion of those passage estimates is reported separately (Table 4) and is informative in its own right, being small for compositionally uniform books and large for composites.

What the Signal Supports

Table 4. Leave-one-book-out performance. Dates BCE; error is true minus estimated, so a positive error means the model placed the book too late. “Pass. SD” is the standard deviation of the book’s passage-level estimates, a measure of internal homogeneity. Every quantity is out-of-sample: standardization, weighting, λ selection and calibration were performed without the held-out book.

Unit	True	Est.	Err.	Pass. SD	$P(\text{post-ex.})$
Amos	760	623	+137	40	0.44
Hosea	745	563	+182	85	0.60
Micah	725	626	+99	67	0.44
Isaiah_1	720	439	+281	95	0.84
Nahum	650	684	-34	0	0.28
Zephaniah	630	661	-31	118	0.32
Habakkuk	605	685	-80	43	0.28
Jer_oracle	605	670	-65	80	0.28
Obadiah	585	625	-40	0	0.44
Ezekiel	580	469	+111	154	0.76
Lamentations	570	696	-126	68	0.28
Isaiah_2	545	667	-122	71	0.28
Haggai	520	551	-31	111	0.60
Zechariah_1	518	431	+87	52	0.84
Malachi	460	515	-55	86	0.64
Joel	400	566	-166	109	0.60
Jonah	400	349	+51	10	0.96
Isaiah_3	400	515	-115	72	0.64
Ezra	380	184	+196	55	1.00
Nehemiah	380	188	+192	83	1.00
Zechariah_2	350	544	-194	49	0.60
Chronicles	330	324	+6	110	0.96
Esther	300	171	+129	54	1.00
Ecclesiastes	250	379	-129	120	0.96
Daniel	167	449	-282	104	0.80

Under this protocol the model achieves a mean absolute error of 118 yr, against 137 yr for the constant predictor that assigns every book the corpus mean date. Rank correlation between true and estimated date is +0.66 ($p = 0.0003$), and 72.9% of the 295 book pairs are placed in the correct chronological order.

However, the same corpus was used to choose the passage size and feature set, so parametric p -values would be optimistic. Significance is therefore established by a permutation null in which book dates are shuffled and the *entire* pipeline is re-run — weighting, λ selection, variance matching and all — 300 times. Permutation is at book level, not passage level, so the clustering structure is preserved under the null; a passage-level shuffle would break the correlation between passages of the same book and produce a null that is far too easy to beat. Against this null the observed rank correlation of +0.66 has $p = 0.003$ (null median

−0.03), the pairwise ordering accuracy of 72.9% has $p = 0.013$ (null median 50.0%), and the mean absolute error of 118 yr has $p = 0.003$ (null median 179 yr).

One arithmetic point deserves emphasis, since pairwise ordering accuracy is commonly reported in this literature without it. The 295 pairs formed from 25 books are not independent, as each book appears in 25−1 of them. Treating the accuracy as a binomial proportion over 295 independent trials gives p -values wrong by ten orders of magnitude. The permutation figures above are the correct ones, and any comparison with pairwise accuracies reported elsewhere should take this into account.

Genre Is the Dominant Confound

The rank correlation just reported is not what it appears to be, and the correction is large enough that reporting the raw figure alone would be misleading.

The anchored corpus is 17 prophetic books, 5 narrative books, and one book each of poetry, wisdom and apocalyptic. Residuals are strongly structured by genre: narrative books are placed +115 yr from their true dates on average and prophetic books −2 yr. 77% of the variance in the model’s predictions lies between genres rather than within them.

This could be an artifact of where the genres sit on the timeline, since all 5 narrative anchors are post-exilic. The test that separates the two is a date-matched contrast. Among anchors written between 470 and 250 BCE, where both genres are represented, narrative residuals average +115 yr and prophetic residuals −133 yr, a difference of +247 yr (Welch $t = 5.08$, $p = 0.0014$). Regressing residual on true date across all narrative and prophetic anchors, the narrative indicator is +293 yr ($t = 5.21$, $p = 0.0001$). The effect is genre, not the endpoint.

The consequence is that raw rank correlation overstates the chronological signal. Centering both truth and prediction within genre, the correlation falls from +0.666 to +0.405. Within the 17 prophetic books alone—a single register spanning 760 to 350 BCE, the best-controlled comparison this corpus permits—it is +0.380 ($p = 0.13$). Both figures are reported throughout in preference to the raw one.

Whether anything survives the confound is a question a permutation null can answer, provided the null is the right one. A null that shuffles dates freely destroys the genre–date association along with the chronology, and so tests only whether the model can tell genres apart, which it plainly can. The appropriate null is restricted: dates are permuted only *within* genre, so the genre–date association this corpus happens to contain is preserved exactly and only the within-genre chronology is destroyed. Against 300 such permutations, each a complete re-run of the pipeline, the observed genre-controlled correlation of +0.405 has $p = 0.017$ (null median −0.077). A chronological signal survives the confound, and it is that signal, not the raw correlation, that the results below rest on.

The Signal Is Lexical, Not Morphosyntactic

The features fall into two families: 250 lexeme frequencies, and 328 morphosyntactic features covering verb stem and tense, clause type and relation, phrase structure, agreement and suffixation. Refitting on each family separately settles which carries the result, and the answer is one-sided.

On the morphosyntactic features alone, genre-controlled ρ is +0.140 and within-prophecy ρ is +0.059. On the lexemes alone it is +0.458 and +0.429, marginally better than the full feature set. Grammatical change over this period, as encoded in the BHSA annotation and measured across 25 books, is not detectable by this model. Whatever diachrony the model uses is vocabulary.

That is not by itself an objection. Lexical replacement is the backbone of the existing method [1], and Persian loanwords and Aramaisms are its standard evidence. But it invites a sharper one: a model keyed to vocabulary may be detecting subject matter rather than date, and the highest-leverage lexemes include *Israel*, *Jacob*, *YHWH* and *holy*.

Two checks argue against that reading. First, high-leverage lexemes are only weakly more genre-diagnostic than low-leverage ones—across the 250 lexemes, leverage correlates with genre η^2 at +0.124, and the fifty highest-leverage lexemes average 0.114 against 0.072 for the fifty lowest. Topic discrimination is too weak to account for a signal this size.

Second, and more directly: the model recovers the textbook first-person pronoun shift unprompted. Of the recognized CBH/LBH diagnostic lexemes, only 2 of 19 are frequent enough to survive the frequency filter—the Persian loans and Aramaisms that constitute the strongest evidence in the literature are all too rare—and both that do survive rank in the top fifth by leverage, at a median rank of 41 of 250. They are the two members of one pair. Classical *ʾanōkî* ranks 32 and pushes estimates earlier; late *ʾānî* ranks 4 and pushes them later. That is the direction the literature predicts, learned from dated books without any encoding of the pair. A topic classifier has no reason to place those two words high and in opposition.

Correcting for the Configuration Search

The figures above come from a configuration — passage size, weighting exponent, ridge penalty — that was itself chosen by comparing out-of-sample performance across a search on this same corpus. A permutation null that holds the winning configuration fixed does not account for that choice, so the reported correlation is the maximum over a search and is optimistic by an unknown amount.

The amount can be measured by nesting the search. For each held-out book, the entire configuration was selected by an inner leave-one-book-out loop over the remaining 24 units, drawn from a grid of 36 combinations, and only then was the held-out book predicted. Nothing about the configuration is informed by the book being scored.

Under that protocol the mean absolute error is 124 yr against 137 yr for the constant predictor, the rank correlation is +0.62 ($p = 0.0009$), pairwise ordering is 71.5%, and 18 of 25 books fall on the correct side of the exile. Selection therefore inflates the rank correlation by 0.04 and the pairwise accuracy by 1.4 points. The attenuated figures are the ones a fresh corpus should be expected to reproduce, and the substantive conclusions do not depend on the difference. The search is also stable: the same weighting exponent is chosen in 24 of the 25 folds, and the passage size alternates between the two adjacent settings whose performance the sweep found indistinguishable.

Variance matching is excluded from this grid deliberately. It is a positive linear map of the predictions, so it leaves both the rank correlation and the pairwise ordering exactly unchanged and cannot inflate either. It is applied always, as an a priori correction for shrinkage of the kind long used in inverse calibration [21], rather than tuned for performance.

Two-Sidedness and Calibration

What matters for the Pentateuchal application is whether the model can say “early” as well as “late”. After inverse-density weighting and variance matching, 6 of the 8 genuinely pre-exilic books are placed before 586 BCE, and 14 of the 17 post-exilic ones after it, giving 20 of 25 overall (Table 4). Without those two steps the model places almost nothing before the exile, and its apparent confidence that everything is late is an artifact of how the corpus is distributed in time rather than evidence about any text.

Fig 1 shows estimated against true date with the jackknife+ bands. Empirical coverage of the nominal 68% band is slightly conservative, which is the expected behavior of jackknife+ intervals at this sample size.

What the Signal Does Not Support

Three limits ought to be stated in the same breath as the results.

First, 118 yr against a 137 yr baseline is a real improvement but a modest one. What the model provides is a chronological *ordering* that has been calibrated onto an absolute scale, not a precise clock. The reported intervals of ± 137 yr at 68% express as much.

Second, the largest single error is Daniel at 282 yr. Genre drives most of this, as narrative books are placed late and poetic books early relative to their true dates (see below). Any single estimate for a text of unusual genre should be read accordingly.

Third, nothing here resolves dates *within* the post-exilic period. A text estimated at 380 BCE with a ± 137 yr interval cannot be distinguished from one estimated at 300 BCE. The claims that follow are therefore confined to the pre-exilic/post-exilic distinction and to relative ordering, both of which the validation supports.

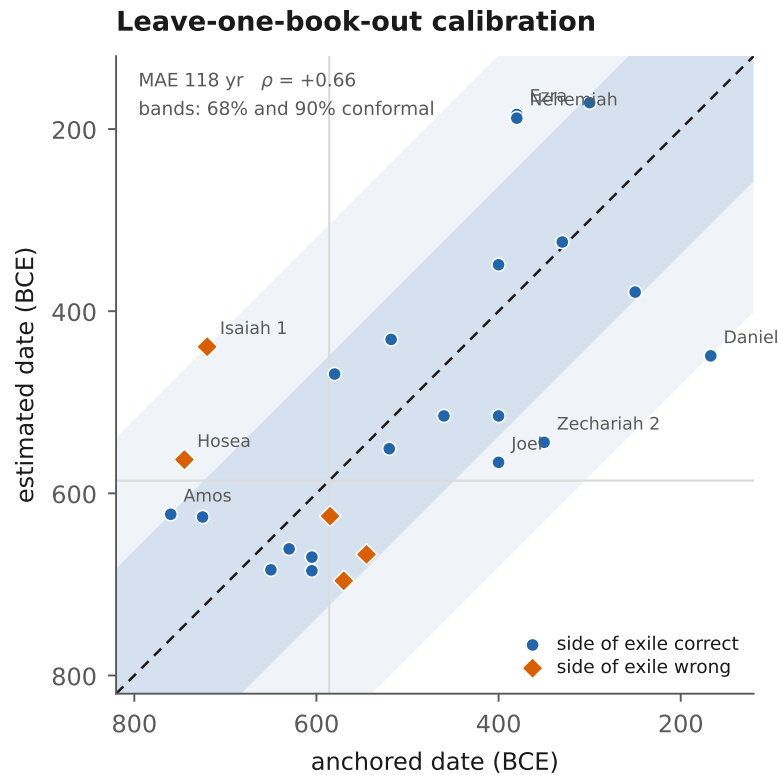


Fig 1. Leave-one-book-out calibration. Each point is one anchored book, dated by a model fitted without it. The dashed line is exact recovery; shaded bands are the 68% and 90% jackknife+ intervals. Books whose estimate falls on the wrong side of the exile are marked with diamonds. Standardization, weighting, regularization and interval calibration were all performed inside the fold.

Results: Within the Sources

The model, fitted on all 25 anchored books, was applied to 19 undated units: the three classical documentary sources, their conventional sub-strata, the five Pentateuchal books as received, the Deuteronomistic prose of Jeremiah, and the three archaic poems. Source partitions follow the standard critical division at chapter granularity, specified in S1 Appendix. Chapter granularity is exact for the sub-strata, whose ranges partition their parent sources without remainder, and nearly so for the three sources: Genesis 6 and 25, where the Priestly and JE material is interwoven verse by verse, are the only chapters assigned to two sources. No target contributed to training, feature selection, weighting, regularization or calibration.

This section asks what the model sees inside a source rather than where it places the source as a whole. Two quantities are available. Dispersion—how widely a source’s own passages scatter—is invariant to the calibration, since variance matching multiplies every prediction by one positive constant and so leaves the ordering of units by dispersion unchanged. Separation between a source’s conventional blocks is not invariant in that way, and turns out not to be invariant to the genre control either. The first survives in part and the second does not, and both are reported.

The Sources Differ in How Internally Varied They Are

Each source is made of many 500-word passages, and each passage receives its own estimate (Fig 2, Table 5). If a source is a single composition, its passages should scatter about as widely as the passages of a single dated book do. If it is a rolling document—an older core with later accretion—it should scatter more.

The yardstick is the anchored corpus itself. Across the 17 anchor books with at least four passages, the within-book standard deviation of passage estimates has a median of 79 yr. That figure is generous to the

sources rather than harsh on them, because several anchors are themselves composite; Isaiah 1–39 is the obvious case.

Dispersion rises with the number of passages a unit contains ($\rho = +0.56$ across the anchors, $p = 0.02$), and the three sources contain 74, 39 and 106 passages, so the comparison is made at matched count. Subsampling each source to 39 passages, the standard deviation of passage estimates is 118 yr for the JE composite, 96 yr for P and 76 yr for D. The ordering is JE > P > D. Bootstrapping over passages, $P(\text{SD}_{JE} > \text{SD}_D) = 1.000$, $P(\text{SD}_{JE} > \text{SD}_P) = 0.935$ and $P(\text{SD}_D > \text{SD}_P) = 0.012$.

Only part of that ordering is robust. Applying the genre screen described in the Validation section, D remains the least dispersed of the three at every screen level, with $P(\text{SD}_{JE} > \text{SD}_D)$ never falling below 0.993. A source usually described as layered is, by this measure, the tidiest one in the Pentateuch, and that holds however hard the genre-diagnostic features are screened out.

The rank of JE does not hold. Unscreened, JE is the most varied of the three, at the 88th percentile of single dated books. With a quarter of the genre-diagnostic features removed, P overtakes it ($P(\text{SD}_{JE} > \text{SD}_P) = 0.429$), and with half removed the comparison is a coin flip (0.445). Screening also inflates every dispersion—JE rises from 118 to 210 yr and P from 97 to 214—because half as many features make a noisier estimate, so some of this is lost power rather than a changed answer. Either way the evidence does not distinguish JE from P, and the claim that JE is the most heterogeneous of the three sources is not supported. What survives is that D is the least.

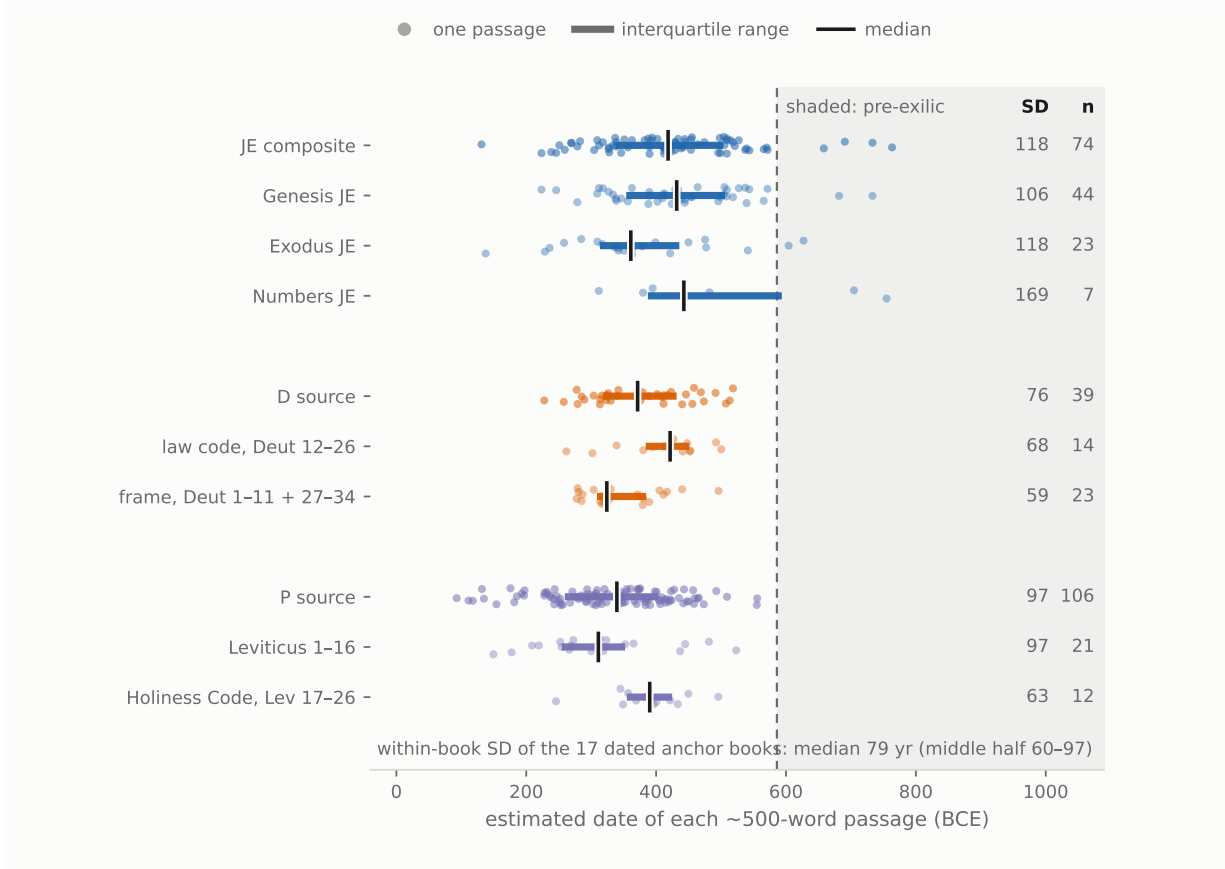


Fig 2. Passage-level estimates within each source. One point per 500-word passage; bars are interquartile ranges and the rule is the median. Sub-blocks are indented beneath their source. The shaded band is pre-exilic. Dispersion within a source and separation between its blocks are different measurements. The separations between blocks shown here do not survive a genre control and are discussed as register effects in the text; the dispersion ordering is robust only for D.

Table 5. Internal structure of the Pentateuchal sources. Dispersion of the estimated dates of the individual ~500-word passages making up each unit, in years BCE. “Pctile” places each unit’s standard deviation in the distribution of within-book dispersions measured on the 17 dated anchor books (median 79 yr): 50% is an ordinary single composition, and higher values are more internally varied than that. Because dispersion rises with passage count ($\rho = +0.56$ across the anchors), the three-way comparison of the sources is made at matched count in the text; the raw values are given here. “Post-exilic” is the share of a unit’s passages estimated later than 586 BCE. Sub-blocks are indented under their source.

Unit	Passages	Median	IQR	SD	Pctile	Post-exilic
JE composite	74	418	166	118	88%	95%
<i>Genesis JE</i>	44	432	152	106	76%	95%
<i>Exodus JE</i>	23	361	122	118	88%	91%
<i>Numbers JE</i>	7	443	206	169	100%	71%
D source	39	372	112	76	35%	100%
<i>law code, Deut 12–26</i>	14	422	67	68	29%	100%
<i>frame, Deut 1–11, 27–34</i>	23	324	76	59	24%	100%
P source	106	340	144	97	71%	100%
<i>Leviticus 1–16</i>	21	311	97	97	71%	100%
<i>Holiness Code, Lev 17–26</i>	12	390	70	63	29%	100%
Jeremiah Dtr prose	30	446	92	89	65%	100%

The Apparent Block Separations Are Register, Not Date

Splitting each source at the division its critical literature already draws produces a striking result that does not survive inspection. I report it, and its failure, because the failure is the more useful of the two.

On the full feature set the blocks separate cleanly and in the expected direction. The law code of Deuteronomy 12–26 is placed +97 yr earlier than the surrounding frame of chapters 1–11 and 27–34 (95% CI +27 to +129, Mann–Whitney $p = 0.007$). The Holiness Code separates from the Priestly material of Leviticus 1–16 by +79 yr ($p = 0.016$), and the JE material of Genesis from that of Exodus by +71 yr ($p = 0.039$). None of those boundaries was supplied to the model, which received chapter ranges and nothing else, and all three point the way two centuries of literary criticism would predict.

The Validation section, however, establishes that genre is the dominant axis of variation in this corpus, and that screening out the most genre-diagnostic features improves the model’s ability to order the anchors once genre is held fixed. Applying that same screen to the block comparisons dissolves them (Table 18). With the most genre-diagnostic quarter of the features removed, the Deuteronomic separation falls to +7 yr ($p = 0.584$) and neither of the others remains significant. With half removed, at the screen level that gives the *best* genre-controlled ordering ($\rho = +0.511$ against +0.404 unscreened), the Deuteronomic separation is –156 yr—reversed, and significant in the opposite direction ($p = 0.005$).

The specification curve shows the same thing from the other side: across the 44 validated specifications the law code is placed earlier than the frame in only 15, and the direction tracks the feature screen almost perfectly.

The reading that fits is unwelcome and simple. The law code is legal prose and the frame is hortatory narrative; the Holiness Code and the Priestly material of Leviticus differ likewise in register. The apparent chronological gap between them was carried by the vocabulary that distinguishes those registers, and when that vocabulary is removed the gap goes with it. These are register effects, not dates.

Two things are worth saying about this. The first is that the loss of significance in the Holiness and JE comparisons is partly a loss of power: screening halves the feature count and the intervals widen accordingly. The Deuteronomic reversal is not that—a sign flip significant at $p = 0.005$ is not what attenuation looks like.

The second is that this is what a genre control is for. A study that reported the unscreened numbers, which are individually significant and align with prior expectation, would have produced three publishable claims and three false ones. The screen that the anchored corpus forces on the dating results turns out to be load-bearing here too, and any future attempt to detect compositional strata this way will have to clear the same bar. Detecting internal structure by linguistic date remains a reasonable ambition; 25 anchor books spanning a single religious literature are not enough to do it.

Three Sources of Variance, and Their Sizes

It is worth stating plainly how the uncertainties in this study compare, since they are easily conflated. The model’s own uncertainty is much the largest: the 68% jackknife+ interval has a half-width of roughly 204 yr for each source. Within-source heterogeneity is next, at 76–118 yr. The instability of the estimate across analytic specifications is the smallest of the three, at 44–65 yr (below).

The sources are more internally variable than the analysis is unstable. A single date for a source made of passages spanning a century or more was never a well-defined quantity to estimate, and this is the reason estimates of it move when the specification moves.

Results: Placement Relative to the Exile

Absolute dates are the weaker half of this study, and I report them in the form the evidence supports: not as point estimates but as the range a defensible analysis can produce.

Every Defensible Specification, at Once

Four analytic choices in this pipeline have no principled answer—the passage size, whether to screen genre-diagnostic features and how hard, whether to weight by inverse density, and whether to variance-match. Crossing them gives 48 specifications across 4 passage sizes, each a complete leave-one-book-out run with its own jackknife+ intervals. Reporting one and relegating the rest to an appendix would invite the reader to treat the reported figure as privileged, which it is not. Fig 3 shows all of them, ordered by how well each orders the anchors once genre is held fixed. 44 of 48 clear the validation threshold of $\rho_{\text{genre}} > 0.20$.

Across the 44 validated specifications, the Deuteronomic source falls after 586 BCE in 44 of 44 and the Priestly source in 44 of 44—without exception. The JE composite falls after the exile in 41 of 44. Median estimates are 482 BCE for JE, 439 for D and 416 for P, with full ranges of 392–615, 345–500 and 287–546 BCE. Averaged over validated specifications, the share of the 68% interval falling after the exile is 73% for JE, 89% for D and 91% for P.

The conclusion the evidence supports is therefore that D and P are post-exilic, and that JE is very probably so but not securely: three specifications place it before the exile. The central estimates fall in the Persian period. The Hellenistic era is not excluded—7 of 44 specifications place P after 332 BCE—but it is not where the estimates point, and no validated specification places JE or D that late.

Reading the lower panel of Fig 3 against the upper ones, no analytic choice sorts the estimates: the best-validating specifications give substantially the same answer as the worst. One choice does affect validation itself. Screening out the half of the features that best discriminate genre raises mean genre-controlled ρ from +0.25 to +0.51. Weighting and calibration make no difference to it at all.

Point estimates for every undated unit under a single reference specification, with their intervals, are given in S8 Table and S1 Fig. They are placed there rather than here deliberately: a table of dates invites exactly the reading this section argues against, and the specification curve above is the honest summary of what the model supports.

The relative ordering is JE → D → P, the classical sequence of the documentary hypothesis, recovered from linguistic evidence alone by a model trained only on independently anchored prophetic and post-exilic books. The ordering is better supported than any of the dates. With intervals of 204 yr, the three sources overlap substantially, and the ordering rests on the model’s demonstrated ability to order rather than on separation between point estimates.

The Archaic Poems and an Extraction Artifact

The three poems traditionally regarded as the oldest Hebrew verse [10, 25, 26] needed a correction before they could be dated at all.

Poems in this corpus are conventionally identified by chapter, but the chapters containing them also contain prose. Exodus 15 is 47% prose by word count, and Deuteronomy 32 is 26%. Because narrative is systematically placed late by this model, including the prose frame drags the poem’s estimate later. The

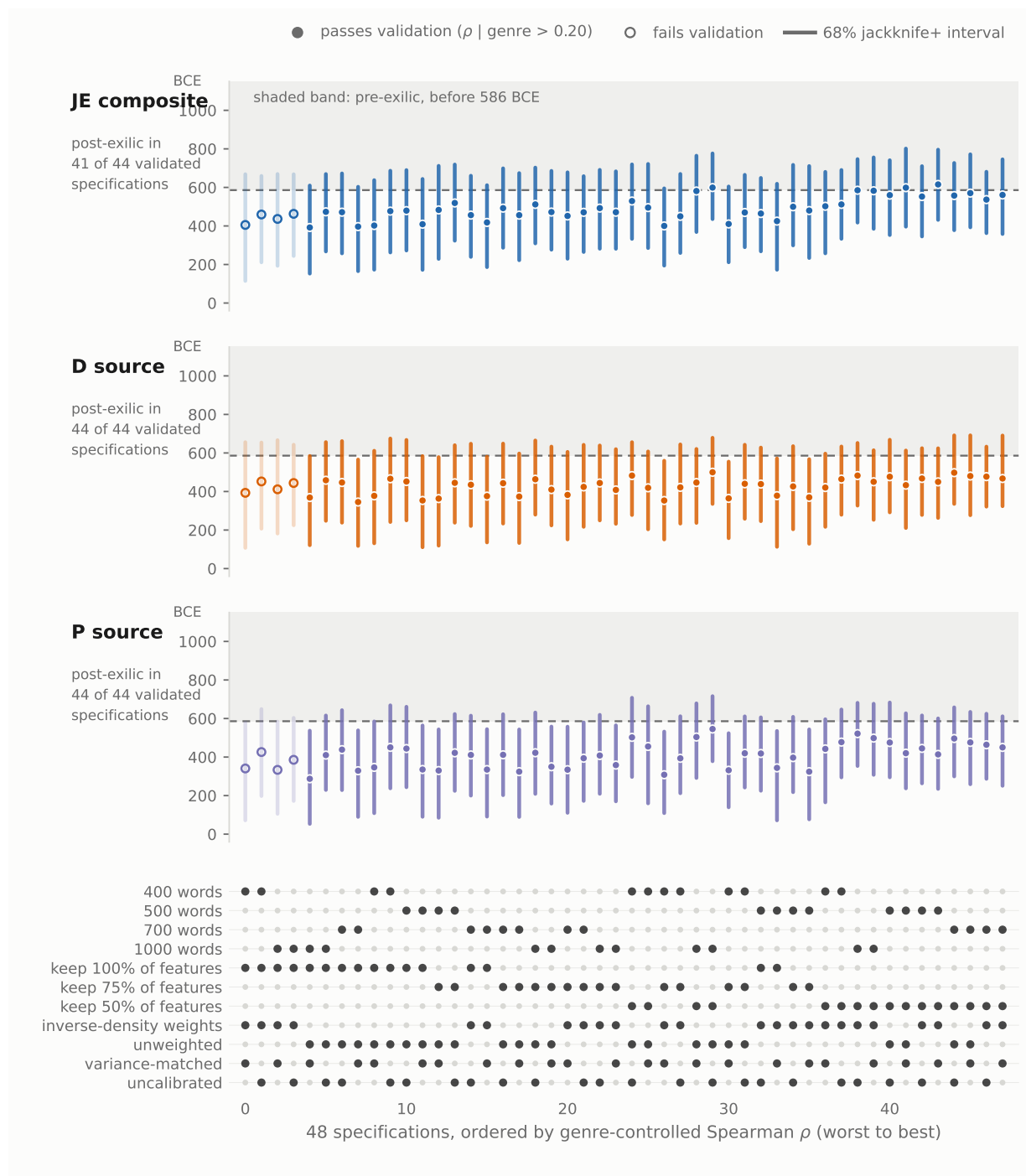


Fig 3. All 48 specifications. Point estimates with 68% jackknife+ intervals for the three sources under every combination of passage size, feature screen, weighting and calibration, ordered by genre-controlled Spearman ρ . Open circles fail validation. The lower panel marks which choices are active in each specification. The intervals never narrow, and no analytic choice organizes the estimates.

effect is not small. The Song of the Sea moves from 638 BCE when extracted as a whole chapter to 780 BCE when extracted at verse precision (Exod 15:1–18). The verse-precise figures are the ones reported here; the chapter-level ones are an artifact, and the convention that produces them is widespread.

At verse precision the Song of the Sea is placed at 780 BCE (1021–560, $P(\text{post-exilic}) = 0.12$), and the Song of Deborah at 819 BCE (1016–585, 0.08). These are the two earliest of the 19 undated units, agreeing with a century of philological argument that they are the oldest surviving Hebrew, and arrived at without any training on them.

The prose continuation of the Song of the Sea (Exod 15:19–27) is placed at 469 BCE, 311 yr later than the poem it follows. A gap of that size between adjacent verses is itself evidence that the poem is not a composition of its narrative frame. Whether such a gap could have been manufactured is taken up below.

The Song of Moses (Deut 32:1–43) is placed at 600 BCE (796–375, 0.50), appreciably later than the other two and straddling the exile. A reading of its contents would not predict this. Its divine-council theology, in which Elyon apportions the nations among the gods and YHWH receives Israel as his share (32:8–9), is usually taken as among the most archaic material in the Hebrew Bible. But the model sees the language rather than the theology, and the language is late. Whether the archaic theology of these verses was preserved in a later idiom, or whether the linguistic estimate is simply wrong for a text this short, cannot be settled from the present evidence.

Applying the Genre Correction to Everything

The genre effect established in the Validation section bears on every estimate in this one, and it must be applied symmetrically or not at all. A correction imposed only where it strengthens a conclusion is not a correction.

The clearest single case is Lamentations, whose date is fixed by its subject matter—it laments the destruction of 586 BCE—making it as securely anchored as any text in the corpus. The model places it at 696 BCE against a true 570 BCE, an error of 126 yr toward greater antiquity. Lamentations is poetry, and poetry as a class carries a mean residual of -126 yr.

For the poems, correcting in the direction the residual indicates makes the claim weaker, and it survives. Subtracting the full 126 yr poetic residual moves the Song of the Sea from 780 to 654 BCE and the Song of Deborah from 819 to 693 BCE. Both remain pre-exilic, both remain the earliest of the 19 undated units, and their probability of being post-exilic rises only to 0.32 and 0.28. For the Song of the Sea to cross the exile the genre effect would have to be 1.5 times the value observed in Lamentations.

For the sources, the same correction runs the other way, and I report it on the same terms. JE is largely narrative, and narrative carries a residual of $+115$ yr. Applying it moves JE from 425 to 533 BCE, D to 486 and P to 454. The minimum probability of post-exilic composition across the three falls from 0.84 to 0.64. All three remain post-exilic, and the ordering is unchanged, but the margin on JE becomes thin. This is the single largest adjustment any defensible correction makes to the results above, and it is the reason JE is described throughout as probably rather than securely post-exilic.

D and P are largely legal material, a register with no anchor at all in this corpus—the 25 anchors contain no law. Their correction is therefore bounded rather than known: between -2 yr if legal prose behaves like prophecy and $+115$ yr if it behaves like narrative, giving minimum post-exilic probabilities of 0.84 and 0.64 respectively. Both bounds leave D and P after the exile. Extending the anchored corpus to include dated legal material is the obvious way to replace that bound with a measurement, and I know of no candidate that does not import the circularity this paper is arguing against.

The Price of Archaizing

Everything above depends on one premise: that the language of a text reflects when it was written rather than when its author wished to seem to be writing. The size of the gap between those two things is measured here, first by performing archaizing on Hebrew directly, and then by turning to Greek texts whose authors performed it themselves.

Synthetic Archaizing in Hebrew

Take a securely dated *late* book, replace its LBH forms with their CBH counterparts at a controlled rate r , re-extract all 578 features from the modified text, and re-date it with a model trained on the other 24 books.

Since I performed the substitution myself, the ground truth is known exactly.

The substitutions are the classical diagnostics of the CBH/LBH literature, applied in the LBH $\rightarrow$ CBH direction: the relative particle $\text{\textit{\text{še-}}} \rightarrow \text{\textit{\text{ašer}}}$; the first-person pronoun $\text{\textit{\text{anī}}} \rightarrow \text{\textit{\text{ankī}}}$; $\text{\textit{\text{malkūt}}} \rightarrow \text{\textit{\text{mamlākāh}}}$; and the negator $\text{\textit{\text{'ēn}}} \rightarrow \text{\textit{\text{lō'}}$, which changes part of speech as well as lexeme. Substitution is applied at the token level before feature extraction, so every downstream feature — lexeme rates, part-of-speech transitions, phrase types, clause relations — sees the altered text exactly as it would if a scribe had written it that way. The rate r runs from 0 to 1, and at $r = 1$ not one LBH token of these types survives anywhere in the book.

Table 6. What total lexical archaizing buys. Each securely dated late book had its Late Biblical Hebrew forms replaced by Classical counterparts at rate r and was then re-dated by a model trained on the other 24 books. $r = 1$ replaces every eligible token in the book. Dates BCE; a positive shift means the book appears older. Ezra, Nehemiah and Chronicles offer too few eligible tokens for their null results to be informative.

Book	True	$r = 0$	$r = 1$	Shift	Swaps	per 1k w
Chronicles	330	298	296	-1	94	2.7
Esther	300	134	139	+5	42	9.1
Ecclesiastes	250	393	424	+30	142	33.5
Daniel	167	441	463	+22	47	13.7
Ezra	380	145	145	+0	13	3.5
Nehemiah	380	155	150	-5	28	3.6
Mean apparent shift				+9		

Total lexical archaizing moves the estimated date by a mean of +9 yr, with a maximum of +30 yr across the 6 books (Table 6). The response is gradual rather than thresholded, as there is no substitution rate at which the estimate suddenly jumps.

However, three of the six books had almost nothing to substitute, with Ezra offering 13 eligible tokens in the entire book, so their null results carry little information. One cannot archaize a text that already lacks the forms. Among the 3 books with substantial substitution density, the exchange rate is about 1.0 yr of apparent antiquity per substitution per 1000 words. Ecclesiastes at $r = 1$ has 33.5 substitutions per 1000 words, roughly one word in thirty replaced, which is a density no scribe sustains across a book, and it moves +30 yr.

Where the Model's Leverage Lies

The reason for so small a number is visible in the model's coefficients.

Fig 5 gives the change in estimated date produced by a one-standard-deviation shift in each feature. Lexical features carry 47% of the total leverage, but that share is spread across 250 features, none of which is worth more than 12.1 yr per standard deviation. No single word is a lever. The rest of the leverage sits in part-of-speech transition rates, verbal aspect and stem distributions, and clause-relation profiles, which are high-frequency grammatical habits rather than choices of vocabulary.

Moreover, these are the features that the passage-level design forces the model to rely on. Requiring a feature to be measurable within 500 words demotes the rare diagnostic vocabulary, which is the manipulable part of the language, and promotes high-frequency morphology, which is not. The resistance to archaizing documented below is therefore a structural consequence of the passage-level design rather than an additional assumption: any model constrained to features estimable in a few hundred words will rest on the same high-frequency morphology.

This can be put quantitatively. Because the model is linear in standardized features, the smallest coordinated feature shift that displaces an estimate by D years has length $D/\|\beta\|$ in standard deviations. For the 311 yr gap between the Song of the Sea and its prose frame, that comes to 5.0 standard deviations across 578 dimensions. What is required is not a word swap but a systematic re-profiling of a text's morphology and syntax, sustained across the whole passage. As the Greek results below show, this is achievable by authors trained to achieve it.

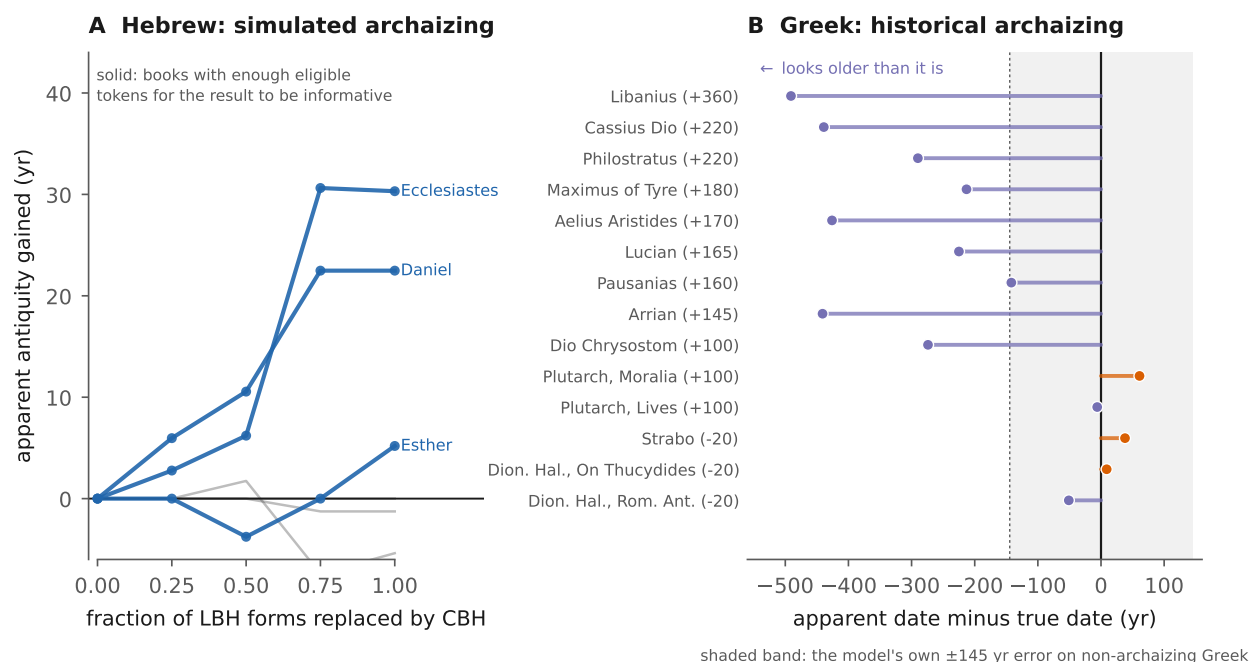


Fig 4. The price of archaizing, by mechanism. **A**, Hebrew: apparent antiquity gained as Late Biblical Hebrew forms are replaced by Classical counterparts at increasing rates in six securely dated late books. Solid lines are the books with enough eligible tokens for the result to be informative. **B**, Greek: displacement of the Second Sophistic Atticizers under a model trained only on non-archaizing Greek, with the model's own error on non-archaizing texts shaded for comparison. Lexical substitution moves an estimate by tens of years; reconstructed grammar moves it by hundreds.

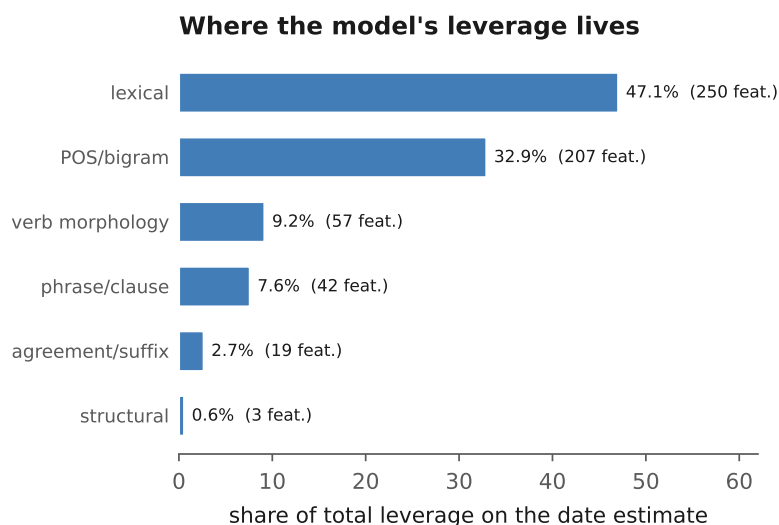


Fig 5. Where the model's leverage lives. Share of total leverage on the date estimate by feature family, where leverage is the absolute change in estimated date produced by a one-standard-deviation shift in a feature, summed within family. Lexical features carry the largest share but spread it across 250 weak features; the majority of leverage sits in morphology and syntax.

Greek Archaizing That Really Happened

The Hebrew experiment simulates archaizing, and it simulates only the lexical component of it. Ancient Greek supplies the real thing, already labeled. During the Second Sophistic, prose authors of the first three

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centuries CE deliberately wrote in imitation of Classical Attic, four to seven centuries after the fact. The program was explicit and institutional. Attic prose composition was taught in the rhetorical schools, dictionaries of permitted Attic usage were compiled, and the imitation succeeded well enough that its practitioners were read as models of style for a millennium. Their dates of composition are securely known from external evidence. This is the labeled set of archaizing texts that Hebrew cannot provide.

The same pipeline was built on a 42-text corpus of non-archaizing Greek prose spanning the Classical and Koine periods, partitioned into 5116 passages, with 747 features. The texts were collected from the Perseus and First1KGreek repositories, where an author’s corpus is stored one work per file, so that the orators, the Lives of Plutarch and the dialogues of Lucian each occupy dozens of separate files. Collecting only the complete corpus of each author matters: an earlier version of this corpus resolved a single work per author and thereby represented Lysias by one speech and Plutarch by one Life, which left the Classical end of the scale, the part the model most needs, severely under-sampled. Greek lacks a morphological annotation comparable to BHSA, so hand-built morphosyntactic rates (optative, infinitive and participle rates, particle and negation rates, the $\tau\tau$ versus $\sigma\sigma$ contrast, sentence length) are supplemented by word-form, character n-gram, and function-word bigram rates, all of them high-frequency by construction as on the Hebrew side. Under leave-one-text-out, the Greek model gives 145 yr against 238 yr for the constant predictor, a rank correlation of +0.69, and 74.7% pairwise ordering accuracy. The replication is therefore a real one, with the same design recovering a diachronic signal in an unrelated language.

The 14 Atticizing texts were excluded from training entirely and then dated (Table 7).

Table 7. Second Sophistic Atticizers under a model trained only on non-archaizing Greek. Dates CE (negative = BCE). A negative shift means the text is dated earlier than it was written, i.e. the archaizing worked. These 14 texts were excluded from training entirely.

Author	Work	True	Est.	Shift	Pass.
Dionysius of Halicarnassus	Roman Antiquities	-20	-71	-51	500
Dionysius of Halicarnassus	Critical essays (15 works)	-20	-11	+9	108
Strabo	Geography	-20	+18	+38	268
Plutarch	Parallel Lives (87 works)	+100	+94	-6	419
Plutarch	Moralia (58 works)	+100	+161	+61	225
Dio Chrysostom	Orations (selected)	+100	-174	-274	308
Arrian	Anabasis of Alexander	+145	-296	-441	140
Pausanias	Description of Greece	+160	+18	-142	223
Lucian	Complete works (71 works)	+165	-60	-225	446
Aelius Aristides	Orations and Sacred Tales (56 works)	+170	-256	-426	540
Maximus of Tyre	Orations (Dialexeis, 5 works)	+180	-33	-213	19
Philostratus	Lives of the Sophists	+220	-70	-290	42
Cassius Dio	Roman History (epitome books)	+220	-219	-439	262
Libanius	Orations (selected)	+360	-131	-491	216
Mean displacement				-206	

The mean displacement is −206 yr, with 11 of the 14 texts placed earlier than they were written (Fig 4B). Libanius is displaced −491 yr. Set against the model’s own 145 yr error on non-archaizing texts, this is a large effect. Deliberate and professionally trained archaizing displaces a text by more than a full error width, and in the worst cases by half a millennium.

The displacement also tracks how programmatic an author’s Atticism was, which is the pattern one would expect if it measures imitation rather than noise. The strict Atticists are displaced furthest: Libanius by −491 yr, Arrian by 441, Cassius Dio by 439, Aelius Aristides by 426, Philostratus by 290, Dio Chrysostom by 274, and Lucian by 225. The authors who wrote a literary Koine rather than a reconstructed Attic are barely moved at all: Plutarch’s *Lives* by 6 yr, Strabo by 38, and Dionysius’ critical essays by 9. A dose-response relation of this kind across authors whose Atticizing intensity is independently described in the literary histories is difficult to obtain from a model that is merely noisy.

Two features of the Greek corpus bear on how this number should be read.

First, the effect is if anything *understated*, because the training set is contaminated. The corpus manifestly classifies Aelian, Appian, Diogenes Laertius, Plotinus, Porphyry, Iamblichus, and the sophistic novelists as Koine, so all of them were used as non-archaizing training material. Yet every one is described in the

literary-historical record as classicizing, and the model placed them among the largest early displacements in the training set even after being told they were not archaizing. A training set containing archaizers teaches the model that late texts look early, which shrinks the measured displacement of the declared Atticizers. Since the classification is a scholarly judgment rather than a datum, I do not adopt one version of it and defend it. Instead the measurement is reported under three definitions of the training set: the manifest's own labels, a stricter classification that reclassifies the authors named above, and a date-blind variant that discards register labels entirely and trains only on texts composed before 50 CE, prior to the Second Sophistic (S6 Table).

The result holds in sign and grows under the stricter definitions. The mean displacement is -206 yr under the manifest's labels, -297 yr under the stricter classification, and -397 yr under the date-blind variant, with 44 of 47 individual texts across the three placed earlier than they were written. The stricter variant is worth a further remark. It removes eight texts from training, and yet the model's own leave-one-text-out error *falls* from 145 yr to 101 yr while its rank correlation rises from $+0.69$ to $+0.80$. Perhaps this ought to have been expected. Removing the suspected archaizers from the training set makes the date model better, exactly as one would expect were they genuinely mislabeled. Nonetheless, the manifest's own labels are used as the primary specification throughout, since that variant is the most conservative and the only one requiring no judgment on my part.

Second, the two Atticizers displaced least, Strabo and Dionysius of Halicarnassus, are the earliest in the set, writing around the turn of the era when the Atticizing program was still young. The displacement grows with the maturity of the movement, which is what one would expect if it measures imitation rather than noise.

Additionally, text lengths remain uneven, from 19 passages to 540, so a short estimate is correspondingly noisy. The result is not driven by the short texts: weighting each text by its passage count gives -201 yr, and restricting to the 13 texts with at least 40 passages gives -206 yr.

The Greek Measurement Under the Same Genre Control

Genre is the dominant confound in the Hebrew corpus, and the Greek corpus is structured in a way that invites the same problem. Its 42 training texts divide into five genres, and genre is correlated with register: the ancient Attic baseline is oratory-heavy and contains no narrative at all, while the Koine material is dominated by history and narrative. A model that had learned to recognize oratory would look like a model that had learned to recognize early Greek. The Atticizers are spread across genres—six histories, three philosophical works, two orations, two narratives and one scientific text—but that spread is not by itself a defense.

The Greek model turns out to be markedly less genre-confounded than the Hebrew one. 45% of the variance in its predictions lies between genres, against 77% for Hebrew, and rank correlation falls only from $+0.689$ to $+0.571$ when genre is held fixed, against $+0.666$ to $+0.405$. With 42 texts spanning eight centuries rather than 25 books spanning six, there is enough within-genre chronological range for the model to use.

The displacement itself is almost untouched. Scoring each Atticizer against the mean displacement of non-archaizing texts *in its own genre* rather than against the corpus as a whole moves the mean from -206 to -202 yr; the genre correction accounts for 2% of it. The decisive comparison is within genre: Atticizing texts are dated 202 yr earlier than non-archaizing texts of the same genre (95% CI 111 to 288 yr earlier, bootstrap $p < 0.001$), and 4 of the 5 genres point the same way. The single exception is scientific prose, which contributes one Atticizing text.

The archaizing measurement is therefore not a genre effect in disguise. This matters more than a routine robustness check would, because the Hebrew half of the paper argues that genre confounding is severe enough to halve a reported correlation; a Greek result that had not been put through the same test would not deserve to be believed.

What the Two Experiments Establish

The two experiments do not come out the same way, and the asymmetry is the substantive finding—though not in the form of a price comparison.

Simulated *lexical* archaizing in Hebrew displaces an estimate by 9 yr, which is 7% of this model's residual scale and thus indistinguishable from no displacement at all. Real *morphosyntactic* archaizing in Greek displaces one by 206 yr, an order of magnitude above that model's noise floor. One arm returns a measurement and the other returns a non-detection, so the two cannot be divided to yield an exchange rate. What can be said is that the detectable case is the grammatical one, and that this falls where the feature-leverage analysis said it would. Swapping out diagnostic vocabulary moves a handful of the 250 weak lexical features and leaves the other half of the model's leverage untouched. The Atticizers did not swap vocabulary. They reconstructed a grammar, reviving the optative, restoring the $\mu\epsilon\nu \dots \delta\epsilon$ articulation, writing $\tau\tau$ for $\sigma\sigma$, and matching Classical participial density. That is the coordinated multi-dimensional shift which the Hebrew sensitivity analysis identified as the only route to a displacement of centuries, and they managed it because an entire educational apparatus existed to teach them how.

The conclusion is therefore conditional rather than absolute, and it ought to be stated in that form. Archaizing can defeat a model of this class; the Greek case demonstrates it. Whether the cheaper lexical route can do so is not answered here, because the instrument used to ask is not sensitive enough to return an answer either way. What it takes is not a scribe who knows the old words, but one trained to write the old grammar and to sustain it across a whole composition. For any particular text, then, the question is whether such training existed in its milieu.

For the Hebrew Bible this becomes a historical question, not a statistical one, and it is not settled here. The institutional preconditions do differ sharply between the two cases. The Second Sophistic had a documented curriculum of Attic prose composition, lexica of permitted usage, and a literary culture that rewarded successful imitation, whereas no comparable program of Classical Hebrew prose composition is attested for the Second Temple period. The Qumran evidence adduced by the dialectal school [5] is the closest analogue, and it is a good deal thinner. The sectarian scrolls mix CBH forms with Aramaisms rather than sustaining a consistent Classical grammar, which on the evidence here would displace their apparent date by tens of years rather than hundreds.

This bears directly on the archaic poems. An archaizing account of the early placement of the Song of the Sea or the Song of Deborah is not excluded by these results, but it carries a specific and demanding burden. It must posit sustained morphosyntactic imitation, of the kind that took the Second Sophistic a formal curriculum to achieve, rather than the lexical substitution that the CBH/LBH literature describes and that is measured above at 9 yr. No comparable institution is attested for Second Temple Judaism. The burden is therefore a heavy one, and it now has a number attached to it.

Discussion

Limitations

Interval width and what the guarantee covers. Jackknife+ intervals are wide here, averaging 443 yr at the 68% level, and their distribution-free warranty is $1 - 2\alpha$, so the nominal 90% level is guaranteed only to 80%. Empirical coverage on the anchored corpus is 72% and 96%, comfortably above nominal, but that is an observation about this corpus rather than a guarantee. No claim in this paper depends on interval width: every substantive claim is a side-of-the-exile claim, an ordering claim, or a claim about dispersion, and none rests on where an endpoint falls. Point dates are reported only as medians across specifications and should not be quoted to a century.

Exchangeability, on which the intervals depend. The jackknife+ guarantee holds for exchangeable observations. The unit here is the book, and these 25 books are not obviously exchangeable: 17 are prophetic against 5 narrative, every narrative anchor is post-exilic, and genre displaces an estimate by up to +115 yr. Under exchangeability the coverage floor is $1 - 2\alpha$; how far it degrades when exchangeability fails in this particular way is established neither in the literature nor here. Empirical coverage on the anchored corpus is above nominal (72% and 96%), which is reassuring, but it is an observation about this corpus and not a guarantee. A reader who doubts the exchangeability of prophecy and narrative should treat the intervals as descriptive.

Genre. Genre, not chronology, is the dominant axis of variation in this corpus: 77% of the variance in the predictions lies between genres, and rank correlation with true date falls from +0.666 to +0.405 once

genre is held fixed. A within-genre permutation null establishes that a chronological signal survives ($p = 0.017$), but the surviving signal is roughly half the raw figure, and every number reported here should be read against the genre-controlled version rather than the raw one. With one securely dated poetic book and no dated legal text at all, the genre correction for two of the three sources is a bound rather than a measurement. This is the largest single weakness of the design, and enlarging the anchored corpus in genre rather than in size is what would most improve it.

The signal is lexical only. Refitting on the 328 morphosyntactic features with every lexeme removed gives a genre-controlled ρ of +0.140 and a within-prophecy ρ of +0.059. Whatever grammatical change occurred over this period is not detectable at this sample size, so the results speak to lexical diachrony and to nothing else. A reader who takes CBH/LBH to be centrally a grammatical distinction will find no support for it here, in either direction: absence of detection across 25 books is not absence of the phenomenon.

Anchor dates that are partly linguistic. Six of the 25 anchors rest on literary rather than external grounds, and two of those (Jonah, Ecclesiastes) are dated partly by the linguistic profile this model measures. Refitting on externally anchored books alone drops genre-controlled ρ to +0.052. That test is confounded—four of the six dropped books are late prophetic, so it also strips the late end of the within-genre range—and the two effects cannot be separated with this corpus. The circularity is reduced relative to conventional practice but not eliminated.

Corpus asymmetry. The anchored corpus is skewed post-exilic, because that is where external synchronisms are dense. Inverse-density weighting mitigates but does not remove the skew, and the pre-exilic end of the scale rests on 8 books. Still, extending the anchored corpus at the early end would do more for this model than any change of estimator.

The two archaizing measurements bound different things. Four substitution pairs is a small vocabulary of archaizing, so the Hebrew experiment prices only the lexical mechanism, and prices it at the extreme setting of total replacement. The Greek experiment prices real imitation but only at the intensity Second Sophistic authors actually achieved, in a language whose annotation is coarser than BHSA, and against a training set whose register classification is a scholarly judgment rather than a datum (S6 Table reports the sensitivity to that judgment). Neither experiment prices sustained morphosyntactic archaizing *in Hebrew*, which is the quantity a critic of the Song of the Sea result would most want. Simulating it would require a model of what Classical Hebrew grammar an imitator could reconstruct, and no such model exists.

The Greek transfer is an analogy rather than a measurement of Hebrew. That Greek imitators achieved a 206 yr displacement establishes that the deep mechanism works in principle and gives its rough scale. It does not establish that a Hebrew scribe could have done the same, and the claim that Second Temple Judaism lacked the institutional apparatus is a historical one, asserted here rather than demonstrated.

One database, one text tradition. All Hebrew results depend on the BHSA annotation of the Masoretic text. Consonantal variants in the Qumran and Septuagint traditions are not represented, and the Masoretic vocalization reflects a much later stage of the language than any composition date discussed here. The features used are drawn from the consonantal text and its morphological parse, but the parse itself is a modern scholarly product.

The Hebrew Experiment Is a Non-Detection, Not a Measurement

The Hebrew archaizing result is reported above as a small number, and it is worth being explicit about what a small number means when the instrument measuring it is this coarse.

Total lexical archaizing of six securely dated late books moves their estimates by 9 yr. The model's own leave-one-out residuals average 127 yr, and the 68% jackknife+ interval on a single undated unit has a half-width of about 230 yr. The observed displacement is therefore 7% of the noise against which it would have to be seen. A shift that size is not distinguishable from no shift at all.

The correct statement is consequently weaker than “lexical archaizing is cheap”, and it is a statement about the instrument rather than about scribes: *this model cannot detect archaizing of this kind*. It does not follow that a Persian-period scribe substituting Classical vocabulary would gain little; it follows that if he did, this model would not notice. The two propositions are easy to conflate and only the second is supported here.

Two features of the design explain the non-detection and bound what it can be generalized to. The substitution operates on the recognized CBH/LBH diagnostic pairs, of which only 2 of 19 are frequent

enough to enter the feature set at all; the rest—the Persian loanwords and Aramaisms that carry most of the weight in the philological literature—are too rare to survive a filter that requires a feature to be measurable within 500 words. And the model’s own signal, though entirely lexical, is spread thinly across 250 common lexemes of which none is worth more than 12.1 yr per standard deviation. Substituting a handful of diagnostic items therefore alters a handful of features out of 250 and moves the estimate very little. A human philologist reasoning from exactly the rare markers this model discards is not subject to the same insensitivity, and may well be harder to deceive than the model is.

The Greek experiment is what carries the argument about archaizing, because there the imitation is real, its authors are known, and the displacement is 206 yr—an order of magnitude above this model’s noise floor, and 202 yr when each Atticizer is compared only with texts of its own genre.

What This Means for the Archaic Poems

The non-detection bears directly on the results most likely to be cited from this paper, and it cuts against them.

The Song of the Sea is placed at 780 BCE and the Song of Deborah at 819. Neither is anywhere near the twelfth- or eleventh-century dates sometimes proposed for them on prosodic and orthographic grounds [25,26]; on this evidence they are early monarchic at best. That gap is worth stating plainly, because a reader who takes these results as corroborating a very early date for the Song of the Sea will have read them backwards.

Set beside that, the poems sit closer to the exile than the point estimates suggest once two displacements are admitted. Lamentations, the corpus’s only securely dated poetry, is placed 126 yr earlier than it was written. Lamentations is not archaizing: it is genuine poetry of its own date, and the displacement is what poetic register alone buys. Applying that correction moves the Song of the Sea to 654 BCE, leaving +68 yr between it and the exile—less than one further correction of the same size.

A deliberately archaizing poem would be expected to exceed that figure rather than match it, since it would carry intentional archaism on top of the register effect Lamentations already exhibits. The displacement required to bring the Song of the Sea to the exile is 1.5 times the Lamentations residual, and to the end of the exile, where the material of Isaiah 40–55 belongs, 1.9 times. Neither multiple is extravagant for a text composed in conscious imitation of older verse.

I do not argue that case here; the evidence assembled in this paper cannot settle it, and the poems’ pre-exilic placement remains this study’s best estimate. What the paper can establish is the shape of the argument that would be needed. An exilic Song of the Sea is not excluded by these results—it requires a register-plus-archaizing displacement of roughly twice what an unremarkable poetic book already shows, in a model demonstrably unable to detect archaizing at all. Whether the poem’s language, imagery and theology support that reading is a question for philology rather than for this model, and one I intend to take up separately.

Which Analyses Were Planned, and Which Were Not

This study grew by responding to its own failures, and a reader is entitled to know which results were sought and which were stumbled into, because the two carry different evidential weight.

Planned before any result was seen: the corpus and its anchoring, the passage-level design, the leave-one-book-out protocol, the free-shuffle permutation null, the archaizing simulation, and the Greek replication. Those are confirmatory.

Everything else in this article was prompted by something that had already gone wrong. The genre analysis began when narrative books were noticed to be systematically misplaced; the within-genre null, the feature screen and the genre-controlled statistic all follow from it. The lexical-versus-morphosyntactic ablation was a response to the suspicion that the model was classifying subject matter. The specification curve was built after the point estimates proved unstable. The block-separation analysis was undertaken as a positive result and is reported as a negative one. The feature-encoding comparison came last of all. These are exploratory, and the two nulls are the only inferential statements among them that were computed against a pre-declared number of draws.

The distinction matters most for the block separations, where an exploratory analysis produced three significant results in the expected direction that a subsequent control dissolved. Had the control not been applied, three false claims would have entered the literature with p -values attached. I report the sequence rather than the tidied outcome so that the reader can weigh the exploratory results accordingly.

Scholarly Context for a Late Torah

The finding that the Pentateuchal sources are post-exilic is not novel in biblical scholarship, however contested it remains. A substantial tradition places the final composition of P in the Persian period [27, 28], and the “Persian period Pentateuch” position associated with the Copenhagen and Sheffield schools places the whole compositional process there [29, 30]. Against this, Hurvitz’s linguistic argument for a pre-exilic P [1, 9] and the broader neo-documentary position maintain earlier dates.

What is offered here is not the direction of the result but its arrival with a calibrated uncertainty, an explicit genre control, and a stated limit on what can be said about archaizing. The linguistic argument for a pre-exilic P has always had to assume that the post-exilic appearance of a text cannot be manufactured, while the argument for a post-exilic P has assumed the converse. This study constrains the second assumption and, in Greek, the first; in Hebrew it establishes only that the question is beyond the resolution of a model of this kind, which is itself worth knowing before the assumption is made again.

Feature Encoding, and a Better Way to Build the Features

Two questions about the feature encoding were raised late in this work and are reported because one of them points somewhere useful.

The features are rates per 500 words rather than proportions, so a linear model in them does not imply an impossible value. But if the trajectory of a form over time is logistic, as language change generally is, then a model linear in the raw rate is fitting a curve with a straight line. Refitting on transformed rates confirms the mismatch is real but small: $\log(1 + x)$ improves mean absolute error from 118 to 111 yr and rank correlation from +0.665 to +0.719. It does nothing for the confound that actually limits this study—genre-controlled ρ is +0.393 against +0.405 unscreened—so the reported model is left on the raw scale.

The second question is more consequential. A rate per 500 words measures how often an author reaches for a form at all, which is substantially a matter of genre, mixed together with which of the competing forms he chooses when he does, which is the diachronic signal. Variationist linguistics separates them by dividing within the set of alternants: the share of *’anōkī* among first-person singular pronouns, rather than *’anōkī* per 500 words. Encoded as a share, that single variable’s genre-controlled correlation with date rises from +0.245 to +0.428.

Applied at the scale the corpus currently permits, the effect is larger than one variable would suggest. 7 alternant pairs have both members above the frequency filter, of which 3 are strict linguistic variables and the rest near-synonyms an automatic procedure cannot distinguish. A model built on those 7 share features alone—against 578 rate features—gives a genre-controlled ρ of +0.554, well above the +0.405 of the full rate model, though its raw correlation (+0.576) and error (125 yr) are worse. That is what the theory predicts: shares discard the frequency information that genre drives, keeping only the choice between forms. Jackknifing over books, the share model’s genre-controlled ρ runs +0.464 to +0.725 with a median of +0.551, against +0.309 for the rate model, and beats it in 24 of 25 single-book deletions. Combining both feature sets gives +0.433 at 115 yr.

Two controls establish that this is the encoding rather than an artifact of fitting seven features instead of 578. Refitting on the *same fourteen lexemes* as plain rates—identical linguistic material, differently encoded—gives a genre-controlled ρ of +0.239, less than half the share encoding’s +0.554. And seven rate features drawn at random give a median of +0.084 over 40 draws, reaching at most +0.500; none attains the share model’s value. Parsimony does not produce this, and neither does the choice of lexemes, since the same-material control holds those fixed.

The pair list, however, will not bear the weight the comparison invites, and the reason is worth reporting because it is the same failure the paper documents elsewhere. Only 2 of the 7 surviving pairs come from the diachronic literature—the first-person pronoun [2, 3] and the *’el/le* merger [3]. The remaining five were

assembled by me as plausible alternations, and three of them are not linguistic variables at all but near-synonyms. Fitted on the two literature-derived pairs alone the model gives -0.213 ; on the five investigator-added pairs it gives $+0.444$. The result does not come from the pairs the literature vouches for.

Nor is it stable across the pair set. Dropping one pair at a time moves genre-controlled ρ between $+0.322$ and $+0.660$, and the seven-pair model is one of 127 non-empty subsets that could have been reported. The number quoted above is therefore a point in a space that was searched after the outcome was visible, and the specification-curve discipline applied to the dating results has not been applied to it.

What survives is narrower and worth keeping separate from what does not. The *encoding* effect is controlled: identical lexemes, differently encoded, differ by more than a factor of two, and no random seven-feature model approaches the share model. The particular seven-pair *feature set* is not a result, and I do not offer it as one. A model built on shares would need its pair list specified in advance on linguistic grounds, and its own passage through the specification curve, the permutation nulls and the ablations—which is a study, not a revision. The reason to say so plainly is that this encoding is the only change tried here that moved the confound rather than the accuracy, and that makes it exactly the kind of promising result that a post-hoc feature list would be most likely to overstate.

Extensibility

Given a morphologically annotated corpus and a set of externally anchored dates, the pipeline is indifferent to the language. The Greek replication was implemented from the same code with a different feature extractor. The binding constraint is annotation rather than method. Ugaritic, Phoenician, and epigraphic Hebrew would supply badly needed early anchors, but they lack comparable morphological parsing. Producing such a parse for the Iron Age epigraphic corpus would be the single most valuable contribution to this line of work.

Relation to Authorship Attribution

Attribution and dating are complementary. A method that assigns a text to a scribal school and a method that assigns it a date constrain different axes of the same question, and their errors are not the same errors. One natural extension would be to use attribution output as a register covariate in the date model, which would let the genre confound documented above be modeled rather than merely reported.

Conclusion

Biblical Hebrew carries a diachronic signal strong enough to order books in time and to separate pre-exilic from post-exilic composition, and far too weak to date any single text to a century. Across 44 defensible specifications the Deuteronomistic and Priestly sources fall after the exile without exception and the JE composite in 41, in the classical sequence $JE \rightarrow D \rightarrow P$, with central estimates in the Persian period. The Song of the Sea and the Song of Deborah remain the earliest of the 19 undated units. No individual date in this study should be quoted to a century, and the intervals say so.

Two findings about method matter more than the dates, and both cut against current practice. Genre, not chronology, is the dominant axis of variation in a corpus assembled this way; a study reporting raw rank correlation without controlling for it overstates its signal by roughly half. And the diachronic signal that survives is entirely lexical: refitting on verb morphology, clause structure and agreement gives nothing. That is a quantitative datum in a debate conducted largely by example, and it favors the lexical emphasis of Hurvitz's method over accounts appealing to grammatical change.

The paper also reports a claim it could not sustain, and I think that is the most instructive part of it. Splitting each source at the division its critical literature already draws produces three significant separations, all in the expected direction, from a model that was told nothing about those divisions. Under the genre control the separations disappear and the Deuteronomistic one reverses. Legal prose and hortatory narrative differ in vocabulary, and that difference was doing the work. The unscreened result is precisely what a study would publish if it stopped one step short of its own control, and it would have been wrong. Detecting compositional strata by linguistic date remains a reasonable ambition; 25 anchor books spanning a single religious literature are not enough to do it, and any future attempt will have to clear the same bar.

The premise on which all such conclusions rest, that a late author cannot convincingly write early, has been treated as an assumption for two decades. It is measurable, and measuring it shows that it is not one assumption but two, with different answers. Complete lexical archaizing of a securely dated late Hebrew book buys a mean of 9 yr of apparent antiquity, since the diagnostic vocabulary that the CBH/LBH literature treats as the substance of archaizing turns out to carry very little of a passage-level model's leverage. The Second Sophistic Atticizers, who reconstructed Classical grammar rather than Classical vocabulary and were trained by a formal curriculum to do so, are pushed 206 yr too early.

Archaizing therefore has a price, and that price is set by how deep the imitation reaches. This is a more useful result than immunity would have been, as it converts a rhetorical impasse into a comparison that can be made case by case. For any text whose date is disputed on linguistic grounds, the question is no longer whether archaizing is possible in the abstract, but whether the milieu in question supported the deep kind of it. For the Second Sophistic the answer is documented and affirmative. For Second Temple Judaism no comparable program is attested, and the closest analogue, the Qumran sectarian scrolls that mix Classical forms with Aramaisms rather than sustaining a Classical grammar, resembles the cheap mechanism rather than the expensive one.

Supporting Information

Table 8. S1 Appendix. Source partitions. Chapter specification of every undated unit. The five Pentateuchal books as received are omitted (they are simply the whole book). Partitions follow the standard critical division; the three poems were additionally re-extracted at verse precision.

Unit	Chapters
P source	Genesis 1, 5, 6, 9, 11, 17, 23, 25, 35, 36, 46, 48; Exodus 6–7, 12, 16, 25–31, 35–40; Leviticus 1–27; Numbers 1–10, 13–20, 25–36
JE composite	Genesis 2–4, 6–8, 12–16, 18–22, 24–34, 37–45, 49–50; Exodus 1–5, 8–11, 13–14, 17–24, 32–34; Numbers 11–12, 21–24
D source	Deuteronomy 1–34
Genesis JE	Genesis 2–4, 6–8, 12–16, 18–22, 24–34, 37–45, 49–50
Exodus JE	Exodus 1–5, 8–11, 13–14, 17–24, 32–34
Numbers JE	Numbers 11–12, 21–24
D law code	Deuteronomy 12–26
D frame	Deuteronomy 1–11, 27–31, 33–34
Song of Moses	Deuteronomy 32 <i>Deut 32:1–43 at verse precision</i>
Holiness Code	Leviticus 17–26
Leviticus P	Leviticus 1–16
Song of the Sea	Exodus 15 <i>Exod 15:1–18 at verse precision (see text)</i>
Song of Deborah	Judges 5 <i>Judg 5:2–31 at verse precision</i>
Jeremiah Dtr prose	Jeremiah 7, 11, 17–18, 21, 24–29, 32–45, 52

Table 9. S2 Table. Effective sample size by passage size. Passages within a book are correlated, so the raw passage count overstates the information available. $n_{\text{eff}} = n / (1 + (\bar{m} - 1)\rho_I)$ with ρ_I the median intraclass correlation across features.

Passage size	Passages	Median ICC	Design effect	n_{eff}	Gain vs $n = 25$
300	470	0.09	2.68	175.3	7.01x
500	285	0.13	2.38	119.6	4.79x
1000	145	0.20	1.96	74.0	2.96x

Table 10. S3 Table. Configuration search. Out-of-sample metrics for each combination of passage size, feature set and calibration under leave-one-book-out. “Coverage” is the empirical coverage of the nominal 68% interval; values near 1.0 indicate conservative intervals and values near 0.5 indicate the shrinkage problem discussed in the Model section.

Passage	Features	p	Calib.	MAE	ρ	Pair %	Coverage
1000	base	579	False	105	+0.56	69.5	0.60
1000	base	579	True	97	+0.56	68.5	0.64
1000	both	1268	False	94	+0.62	71.5	0.58
1000	both	1268	True	96	+0.62	71.9	0.68
1000	ngram	689	False	96	+0.59	70.8	0.53
1000	ngram	689	True	99	+0.60	70.8	0.67

Table 11. S4 Table. Feature leverage by family. Leverage is the shift in apparent date, in years, produced by moving a feature one standard deviation while holding the rest of the passage fixed. No family carries the signal alone and no single feature approaches the size of the effects being estimated; the full per-feature listing follows.

Family	Features	Total leverage	Share
lexical	250	573	47.1%
POS/bigram	207	401	32.9%
verb morphology	57	112	9.2%
phrase/clause	42	93	7.6%
agreement/suffix	19	32	2.7%
structural	3	7	0.6%

Table 12. S4 Table, continued. Per-feature leverage. All 578 features, years of apparent date per standard-deviation shift, ordered by magnitude. Read down the first pair of columns, then the second, then the third.

Feature	yr/sd	Feature	yr/sd	Feature	yr/sd
lex.<MD[	-12.12	prs_M	+2.63	pb.prin_prde	-1.09
vs_hit	-6.88	pb.advb.advb	+2.63	rela_Adju	-1.09
lex.QRB/	+6.69	lex.XRB/	+2.62	typ_VP	+1.08
lex_RB/	-6.56	lex.NTN[	-2.61	lex.MXNH/	-1.08
pb.advb.verb	+6.34	lex.MLK[	-2.61	lex.NXLH/	+1.08
pb.prin.conj	-6.09	pb.verb.nmpr	+2.60	lex.XV>T/	+1.06
pb.prde.verb	-5.97	lex.JCB[	+2.59	lex.JHWDH/	-1.06
lex.>NJ	-5.89	pb.prde.intj	+2.58	vt.infa	+1.06
typ_CP	-5.74	prs_NW	-2.57	pos.inrg	+1.05
pb.intj.art	+5.74	lex.FDH/	+2.57	vt.impv	+1.05
lex.>RY/	+5.68	lex.PR<H/	+2.55	vx.impv.hif	-1.04
pb.art.nmpr	+5.59	pb.conj.advj	-2.54	pb.subs.advb	+1.04
vx.perf.nif	-5.51	vx.impv.nif	+2.54	pb.intj.advb	+1.04
pb.prep.advb	+5.50	vsf.hif	-2.52	vtf.perf	+1.04
lex.>DMH/	+5.40	vx.wayq.hif	-2.51	lex.KP/	+1.04
lex.J<QB/	+5.26	lex.<BR[	+2.50	pb.advb.nmpr	-1.03
lex.CM/	-5.17	lex.LQX[	+2.50	lex.BGD/	-1.03
pos.conj	-5.12	lex.R>C/	+2.50	pb.prde.conj	-1.02
pdp.conj	-5.10	rela.Subj	-2.45	lex.<FR/	-1.01
lex.DBR/	-5.08	pb.verb.art	-2.43	lex.>LP=/	-1.01
lex.JFR>L/	+5.08	lex.>TH	+2.43	lex.>HRN/	-1.00
lex.M>D/	-4.99	pos.intj	+2.43	lex.>X/	-0.99
lex.<WLM/	-4.89	pb.inrg.intj	+2.42	pb.nmpr.prde	-0.98
pb.nega.subs	+4.88	pb.inrg.verb	+2.42	pb.advj.advb	+0.98
pb.inrg.art	-4.83	fun.Adju	+2.42	pb.conj.prps	-0.98
lex.GDWL/	-4.77	lex.CLWM/	-2.40	pb.nmpr.subs	-0.97
pb.nmpr.advj	+4.76	pdp.art	-2.40	nu.pl	+0.97
lex.BW>[	-4.75	lex.TWK/	+2.39	fun.Pred	+0.97
pos.advb	+4.75	pb.advb.prep	-2.39	lex.>M/	+0.96
lex_KH	+4.72	pb.advb.intj	+2.38	lex.JC<[	-0.96
lex.NF>[	+4.72	pb.advj.prep	-2.37	lex.JRD[	+0.96
lex.GM	-4.72	ps.p2	+2.37	lex.>T	-0.95
rela.Objc	-4.71	pdp.intj	+2.36	lex.CKB[	+0.95
fun.Loca	+4.67	pb.verb.nega	+2.36	lex.JKL[	-0.94
pb.subs.conj	-4.66	pb.subs.prin	-2.36	pb.nega.prde	-0.94
lex.XJL/	-4.65	pb.nmpr.art	-2.34	lex.CNJM/	-0.94
lex.JHWH/	+4.62	lex.R<H/	-2.33	pb.prde.subs	-0.94

Feature	yr/sd	Feature	yr/sd	Feature	yr/sd
lex.<M	-4.61	pb_prep_verb	-2.33	lex_RC</	-0.92
pb_nmpr_prin	+4.57	nu_sg	+2.32	lex.>XR/	-0.92
lex.<NH[	+4.56	lex.QRB[	+2.31	lex.MQWM/	+0.92
pb_advb_inrg	+4.55	pb_prps_advb	-2.28	vsf_hof	+0.91
fun_Conj	-4.55	pb_verb_verb	+2.28	pb_verb_inrg	+0.89
lex.Z>T	+4.54	vx_impf_piel	+2.28	vx_ptca_piel	+0.88
pb_nmpr_advb	+4.53	lex.>TM	-2.27	lex.>BN/	+0.88
vsf_hit	-4.52	pb_prep_prep	+2.27	pdp_inrg	+0.88
lex.QDC/	-4.49	vsf_nif	-2.26	pb_nmpr_intj	+0.86
cl_per_sent	-4.49	rela_PreC	-2.26	lex.MDBR/	+0.85
pb_prde_adjv	-4.48	lex.BQR=/	+2.24	rela_RgRc	+0.85
lex.PQD[	+4.48	pb_advb_prde	-2.23	lex.MZBX/	+0.85
lex.LKN	+4.47	pb_prde_art	+2.22	vx_impf_nif	+0.85
vx_impf_hif	+4.45	vsf_poel	+2.22	vs_hof	+0.85
lex.VWB/	-4.45	pb_prps_prep	+2.21	lex.<Y/	-0.84
pb_prps_nmpr	-4.44	lex.GBWL/	+2.21	lex.NKH[	-0.84
lex.MYRJM/	+4.38	lex_H	-2.20	lex_MH	-0.84
pb_prep_neg	-4.35	pos_art	-2.20	lex.CPV[	+0.83
lex.<BD/	-4.35	pb_prde_prin	-2.18	typ_IPrP	-0.83
lex_MLK/	-4.32	pb_advb_conj	+2.18	pb_adjv_art	-0.82
pdp_advb	+4.31	lex_JSP[	+2.17	fun_Subj	-0.82
lex.BFR/	-4.30	vt_ptcp	+2.16	pdp_prps	-0.82
typ_AdvP	+4.27	pb_prin_prep	+2.16	lex.<WD/	+0.82
lex_L	-4.26	pb_subs_prep	+2.15	lex.<WN/	+0.81
lex.>NKJ	+4.17	pb_adjv_intj	+2.14	pb_subs_nmpr	+0.81
lex.>P/	+4.17	lex.<LH[	+2.13	pb_prep_subs	+0.81
prs_H	+4.17	lex.KTB[	-2.13	lex_LXM/	+0.79
lex_RBH[	-4.16	pb_conj_inrg	-2.10	lex_KRT[	+0.77
vx_impf_qal	+4.13	pb_prde_prps	-2.09	pb_inrg_adjv	+0.77
lex.FR/	-4.12	pb_verb_advb	+2.07	typ_NegP	+0.77
pb_prin_neg	+4.09	pb_subs_verb	+2.07	pb_intj_conj	-0.76
lex_JRWCLM/	-4.07	lex.CLMH=/	-2.06	lex_RWX/	+0.76
vtf_impf	+4.06	pb_nmpr_inrg	+2.06	lex_ZKR[	+0.75
lex_NYL[	+4.03	lex.N>	+2.05	lex.>JC/	+0.75
lex_K	+4.01	pb_nmpr_verb	+2.05	lex_R</	+0.75
lex.>KL[	+3.96	ph_per_clause	-2.04	pb_adjv_verb	+0.75
pb_nmpr_neg	+3.94	vt_perf	+2.03	pb_prep_art	-0.74
pb_inrg_nmpr	+3.94	fun_Nega	+1.98	ps_p3	+0.74
lex.R>H[	-3.94	vs_hif	-1.98	prs_HM	+0.73
lex_PNH/	-3.92	pb_art_prde	+1.98	lex_JHWC</	+0.73
fun_Modi	+3.90	lex_KSP/	-1.95	pb_prps_art	+0.73
rela_Attr	-3.89	st_c	+1.94	lex_BT/	+0.72
fun_Rela	-3.88	lex_H=	+1.91	rela_Coor	+0.72
lex.>LH	-3.81	pb_art_verb	+1.91	pb_negart	-0.71
lex_CM	+3.81	vs_poel	+1.91	pb_intj_nmpr	+0.70
pb_prde_neg	+3.79	lex.>DWN/	+1.90	pdp_neg	+0.70
fun_Objc	+3.78	lex.BN/	-1.90	vtf_impv	+0.68
pb_neg_prep	-3.78	pb_nmpr_prep	+1.88	vx_ptca_nif	+0.68
lex.TXT/	-3.76	lex.HJH[	+1.87	lex.MWT[	-0.67
lex.>B/	-3.76	pb_prde_prep	+1.87	pb_adjv_nmpr	-0.66
lex_BD/	-3.75	pos_adjv	-1.87	lex.>W	+0.66
lex_HR/	+3.74	pb_inrg_prep	-1.87	lex.>SP[	+0.66
lex_LB/	-3.73	pb_conj_art	-1.85	pb_prep_nmpr	-0.66
lex_MJM/	+3.73	pb_adjv_adjv	+1.85	pb_intj_intj	+0.66
lex.<T/	-3.68	pb_subs_neg	+1.81	lex.QR>[	-0.63
lex_CNH/	-3.67	pb_prin_advb	-1.81	lex_JY>[	-0.62
vt_wayq	-3.66	pb_intj_prin	+1.80	vtf_inf	+0.62
lex.DWD==/	-3.66	pb_prps_prde	-1.79	lex.>RB</	+0.62
lex_CMN/	+3.66	pb_prep_intj	+1.79	vx_ptca_qal	-0.61
lex.>HB[	+3.65	vtf_inf	-1.79	lex_PLCTJ/	-0.61
pb_intj_adjv	+3.65	lex.KLJ/	-1.77	lex_KWN[	+0.61
fun_Intj	+3.63	vtf_ptcp	+1.75	lex_BHMH/	-0.60
pb_prin_nmpr	+3.61	lex.C>WL=/	-1.75	pb_prde_inrg	-0.58
vtf_wayq	-3.61	lex_XMC/	+1.74	pb_inrg_prps	-0.57
lex_CLC/	-3.58	pb_prep_prin	+1.74	rela_Spec	-0.57
lex.>CR	-3.58	lex.SWR[	+1.72	lex.>DM/	+0.55
lex.N>M/	+3.58	pb_prps_intj	+1.71	rela_Cmpl	-0.55
lex_W	-3.57	pb_prps_conj	+1.71	prs_MW	+0.54
lex.LJLH/	+3.57	pdp_subs	+1.71	typ_PrNP	-0.53
pb_prps_neg	+3.56	lex.M>H/	-1.70	ps_p1	+0.52
prs_W	-3.56	pb_prep_prps	+1.70	pos_prin	-0.51
lex_JR>[	-3.54	clause_len	-1.68	lex.CWB[	-0.51
pb_neg_advb	+3.52	vx_perf_piel	+1.68	vx_impv_piel	+0.51
lex_XZQ[	-3.52	lex.JWM/	+1.67	pb_prps_prin	+0.50
lex_MN	+3.51	lex.<FRJM/	-1.67	vs_piel	+0.49
lex.>C/	+3.50	pb_advb_prps	+1.67	pb_prde_advb	+0.49
lex.<FH[	-3.49	pb_conj_verb	-1.66	lex_XJ/	+0.49

Feature	yr/sd	Feature	yr/sd	Feature	yr/sd
lex.<D	-3.48	lex.CM<[	+1.65	vx_impv_qal	+0.48
lex.>LHJM/	-3.48	lex_PH/	-1.64	pdp_prep	-0.48
fun.Cmpl	-3.48	pos.nega	+1.60	lex_RGL/	+0.48
lex_MW<D/	-3.48	pos.subs	+1.59	st.a	+0.47
vx_wayq_qal	-3.47	lex_CC/	-1.59	pb.verb_prde	-0.46
lex_N<R/	+3.47	pb.subs_prps	+1.58	pb_nmpr_prps	+0.46
pb.subs_prde	-3.45	pb.prps_prps	-1.57	vsf_hsst	+0.45
lex_HMH	+3.45	pos.verb	+1.56	vx_perf_hif	-0.45
lex_BRJT/	-3.45	verb_rate	+1.56	lex.<JR/	+0.44
pb.art_adjv	-3.45	lex_BRK[	-1.55	pb_nmpr_nmpr	+0.44
lex_MCPV/	+3.45	pb.prde_nmpr	+1.55	pb.nega_intj	-0.44
pb.verb_prps	-3.45	typ_PPpP	-1.53	pb.verb_adjv	+0.43
pb_adjv_conj	-3.44	pb.nega_verb	+1.53	lex_JLD[	-0.43
gn.f	+3.42	lex_CMR[	-1.52	lex.TWRH/	+0.41
lex.>L	+3.42	lex_MY>[	-1.51	lex.>L/	+0.41
pb_prps_adjv	-3.40	lex_MNXH/	-1.50	lex_ZH	-0.41
lex_HW>	+3.39	pb_adjv_prde	+1.49	vt_ptca	+0.40
vsf_qal	+3.36	pb.prin_adjv	+1.48	lex_HM	-0.40
vt_impf	+3.35	lex_QWL/	-1.48	pb_prep_prde	-0.40
pb.verb_subs	+3.35	lex.>RWN/	-1.48	pb.prin_verb	+0.38
lex_NVH[	+3.32	pb.advb_adjv	-1.48	lex.>T==	+0.37
lex_KBWD/	+3.30	lex_ZHB/	-1.48	pb_nmpr_conj	-0.37
lex_MCH=/	-3.29	lex_JWSP/	+1.47	pdp.prin	-0.37
pb_inrg_prde	-3.28	pb.subs_intj	-1.47	pb_intj_prps	-0.36
lex_DRK/	+3.26	pb.subs_art	-1.46	fun.PreC	+0.36
lex.<L	+3.24	lex_KJ	+1.45	pb.subs_inrg	+0.35
rela_NA	+3.24	lex.<M/	+1.45	vsf_piel	+0.34
pb.prin_subs	-3.24	pb_inrg_advb	+1.45	pb.prin_art	+0.33
vs_pual	+3.24	lex_XJH[	+1.45	lex_NPL[	-0.31
sent_len	-3.22	pb.advb_art	-1.45	lex_LM<N	-0.31
pb.advb_nega	-3.21	lex_CTH[	+1.44	pb_adjv_prps	-0.31
typ_AdjP	-3.19	lex.<TH	+1.44	lex_JD<[	-0.30
vsf_pual	+3.18	lex.L>	+1.44	pos_prep	-0.29
lex_XYR/	+3.13	pb_adjv_prin	-1.43	lex_NPC/	-0.28
lex_CMJM/	-3.12	vt_infc	-1.43	lex.>JB[	-0.28
lex_KL/	-3.11	pb_intj_nega	-1.43	rela_ReVo	-0.27
lex_C<R/	+3.10	lex_KLH[	-1.41	pb.verb_prin	-0.25
pb.nega_adjv	+3.10	pb.conj_nmpr	-1.41	lex.>L=	+0.24
pb_prep_conj	-3.05	pb.verb_intj	+1.41	lex_JRC[	-0.24
lex_KN	+3.04	fun.Pre0	+1.39	lex_M<FH/	-0.23
vx_ptca_hif	-3.02	pb.verb_prep	-1.38	pb.conj_nega	-0.22
lex_YB>/	+3.02	typ_InrP	+1.38	vs_hsst	+0.22
pb.conj_prep	-3.02	pb.nega_nmpr	+1.38	prs_J	-0.21
vs_qal	+2.99	pb.conj_advb	-1.37	rela_Resu	-0.21
pb_intj_prep	+2.99	pb.subs_adjv	+1.37	pos_nmpr	+0.19
lex_DM/	+2.98	pb_prep_adjv	+1.36	pb.prin_prps	+0.19
lex_DBR[	-2.97	lex.>CH/	-1.33	pb_prps_subs	-0.17
pb.art_prps	+2.96	lex_JM/	+1.33	pdp_prde	+0.17
pb_intj_prde	+2.92	lex_NGD[	-1.32	lex_NBJ>/	+0.16
lex.>HL/	-2.92	pb_prep_inrg	+1.31	pb_intj_verb	+0.16
pb.conj_prde	-2.92	lex_HNH	+1.30	lex_MJ	+0.16
lex_MLXMH/	-2.91	lex_KHN/	+1.30	pdp_nmpr	+0.15
lex_BNH[	-2.89	lex_JD/	-1.27	lex.<ZB[	-0.15
lex.>XD/	-2.88	lex_SBJB/	+1.26	prs_any	+0.15
lex_XV>[	-2.87	pb_prps_verb	+1.25	lex.>M	+0.14
vx_wayq_piel	-2.87	pos_prde	-1.24	pb.verb_conj	+0.14
lex.>DNJ/	+2.87	phrase_len	-1.24	pb_inrg_inrg	+0.13
lex_HLK[	+2.86	typ_PP	-1.23	pb_inrg_nega	+0.13
lex_BJT/	+2.85	pb_adjv_subs	+1.22	lex.<LH/	+0.13
pb.advb_prin	-2.84	pb.conj_prin	-1.21	pb_intj_subs	-0.13
lex_Y>N/	-2.83	prs_K	-1.20	fun.Voct	-0.11
typ_NP	+2.80	lex.<JN/	-1.19	pos_prps	+0.10
typ_InjP	+2.80	pdp.verb	+1.19	pb_adjv_inrg	+0.10
lex.>JN/	-2.78	pb_inrg_subs	+1.19	lex_LWJ/	-0.10
lex_ML>[	+2.77	nu_du	+1.19	pb.conj_subs	-0.09
fun.Time	-2.77	lex_LBB/	-1.18	lex.<BD[	-0.08
pb.art_subs	-2.77	pb.conj_intj	+1.16	lex_ML>K/	-0.08
lex_BQC[	-2.77	lex_BJN/	-1.15	lex_XDC=/	+0.08
lex_QWM[	+2.75	fun.Ques	+1.15	lex_YDJQ/	+0.05
lex_CLX[	+2.75	lex_HJ>	-1.14	pb_inrg_conj	+0.05
gn.m	-2.75	vs_nif	-1.14	pb.conj_conj	-0.05
vx_perf_qal	+2.74	lex_CB</	-1.13	lex.B	+0.04
lex_ZR</	-2.74	vtf_ptca	-1.13	vx_wayq_nif	-0.04
pb_prps_inrg	-2.73	pb.nega_conj	+1.12	pb_adjv_nega	-0.03
pb.nega_prps	-2.72	pdp_adjv	+1.11	lex_MVH/	-0.02
pb.subs_subs	+2.72	lex_GWJ/	-1.11	vs_poal	+0.01
lex.>MR[	+2.70	lex_YWH[	+1.11	vsf_poal	+0.01

Feature	yr/sd	Feature	yr/sd	Feature	yr/sd
lex_>MH=	+2.68	lex_FJM[	-1.11	lex_XSD/	+0.01
pb_advb_subs	+2.64	lex_BBL/	+1.10		

Table 13. S5 Table. The Greek corpus. All 63 texts, with the composition date and its assumed standard deviation, the genre, the register label from the manifest, and the token count after extraction. Dates are CE; negative values are BCE. The fourteen texts labelled Atticizing are the archaizing authors whose displacement is measured in the Greek section; they are excluded from the training set under variant A of Table 14. A dagger marks the five texts reserved as a date-validation holdout.

Author	Work	Date	Genre	Register	Tokens
Herodotus	Histories	-440 ± 20	history	ancient Attic	163 991
Thucydides	History of the Peloponnesian War	-410 ± 15	history	ancient Attic	140 747
Andocides	Orations (complete, 4 works)	-400 ± 20	oratory	ancient Attic	6 935
Lysias	Orations (complete, 34 works)	-390 ± 20	oratory	ancient Attic	36 737
Plato	Republic	-375 ± 20	philosophy	ancient Attic	54 685
Isocrates	Orations and letters (complete, 30 works)	-375 ± 20	oratory	ancient Attic	109 242
Xenophon	Anabasis	-370 ± 15	history	ancient Attic	48 662
Xenophon	Hellenica	-355 ± 15	history	ancient Attic	59 275
Plato	Laws	-350 ± 15	philosophy	ancient Attic	66 846
Demosthenes	Orations (complete Demosthenic corpus, 63 works)	-350 ± 15	oratory	ancient Attic	248 235
Aeneas Tacticus	How to Survive Under Siege (Poliorketika)	-350 ± 20	science	ancient Attic	13 742
Aeschines	Orations (complete, 4 works)	-345 ± 15	oratory	ancient Attic	30 266
Hyperides	Orations (complete, 6 works)	-340 ± 15	oratory	ancient Attic	8 428
Aristotle	Nicomachean Ethics	-335 ± 10	philosophy	ancient Attic	51 252
Aristotle	Politics	-335 ± 10	philosophy	ancient Attic	70 709
Theophrastus	Characteres	-315 ± 15	philosophy	ancient Attic	6 241
Theophrastus	Historia Plantarum	-310 ± 15	science	ancient Attic	66 973
Septuagint (anon. translators)	Genesis (LXX)	-270 ± 20	narrative	LXX	20 136
Septuagint (anon. translators)	Exodus (LXX)	-260 ± 20	narrative	LXX	18 947
Septuagint (anon. translators)	1 Kingdoms / 1 Samuel (LXX)	-240 ± 30	history	LXX	12 250
Polybius	Histories †	-160 ± 25	history	Koine	280 713
Apollodorus (pseudo-)	Bibliotheca (Library of Greek Mythology)	-150 ± 75	history	Koine	23 300
Septuagint / Jewish author	Judith (LXX)	-125 ± 40	narrative	LXX	6 579
Jason of Cyrene (epitomized, anon.)	2 Maccabees (LXX)	-110 ± 25	history	LXX	10 381
Anon. Jewish historian	1 Maccabees (LXX)	-100 ± 25	history	LXX	15 482
Diodorus Siculus	Bibliotheca Historica (Books 1–5, 11–20)	-50 ± 20	history	Koine	175 723
Dionysius of Halicarnassus	Roman Antiquities	-20 ± 15	history	Atticizing	258 387
Dionysius of Halicarnassus	Critical essays (15 works)	-20 ± 15	philosophy	Atticizing	54 710
Strabo	Geography	-20 ± 15	science	Atticizing	137 058
Philo of Alexandria	De Opificio Mundi	40 ± 15	philosophy	Koine	4 706
Philo of Alexandria	De Vita Mosis	40 ± 15	history	Koine	11 644
Anonymous	Vita Aesopi (Life of Aesop)	50 ± 75	narrative	Koine	7 008
Mark (anonymous)	Gospel of Mark †	70 ± 15	narrative	Koine	6 908
Josephus	Jewish War	80 ± 10	history	Koine	118 639
Matthew (anonymous)	Gospel of Matthew †	85 ± 15	narrative	Koine	8 190
Josephus	Jewish Antiquities (selected books)	93 ± 5	history	Koine	131 852

Author	Work	Date	Genre	Register	Tokens
John (anonymous)	Gospel of John	100 ± 20	narrative	Koine	9 117
Chariton of Aphrodisias	Callirhoe (Chaereas and Callirhoe)	100 ± 40	narrative	Koine	12 339
Anonymous (Jewish pseudepigraphon)	Testament of Abraham (Recension A)	100 ± 50	narrative	LXX	4 047
Plutarch	Parallel Lives (87 works)	100 ± 15	history	Atticizing	215 503
Plutarch	Moralia (58 works)	100 ± 15	philosophy	Atticizing	115 215
Dio Chrysostom	Orations (selected)	100 ± 15	oratory	Atticizing	158 221
Ignatius of Antioch	Seven Genuine Letters	108 ± 5	narrative	Koine	5 618
Arrian / Epictetus	Discourses of Epictetus	108 ± 10	philosophy	Koine	63 725
Luke (anonymous)	Gospel of Luke [†]	120 ± 30	narrative	Koine	6 006
Arrian	Anabasis of Alexander	145 ± 10	history	Atticizing	71 742
Achilles Tatius	Leucippe and Clitophon	150 ± 30	narrative	Koine	35 505
Appian	Roman History (selected books)	155 ± 15	history	Koine	108 302
Pausanias	Description of Greece	160 ± 15	history	Atticizing	113 812
Lucian	Complete works (71 works)	165 ± 15	narrative	Atticizing	228 834
Aelius Aristides	Orations and Sacred Tales (56 works)	170 ± 15	narrative	Atticizing	277 975
Maximus of Tyre	Orations (Dialexeis, 5 works)	180 ± 15	philosophy	Atticizing	9 419
Galen	On the Natural Faculties	180 ± 15	science	Koine	182 296
Clement of Alexandria	Stromateis	200 ± 15	philosophy	Koine	26 724
Philostratus	Lives of the Sophists	220 ± 15	history	Atticizing	21 892
Aelian	Varia Historia	220 ± 15	history	Koine	32 245
Cassius Dio	Roman History (epitome books)	220 ± 15	history	Atticizing	134 066
Diogenes Laërtius	Lives of the Eminent Philosophers [†]	230 ± 30	history	Koine	48 271
Plotinus	Enneads (selected)	265 ± 10	philosophy	Koine	78 490
Porphyry	Life of Plotinus	290 ± 15	history	Koine	13 780
Iamblichus	On the Pythagorean Life	305 ± 15	philosophy	Koine	11 113
Eusebius of Caesarea	Ecclesiastical History	315 ± 15	history	Koine	22 313
Libanius	Orations (selected)	360 ± 15	oratory	Atticizing	110 224

Table 14. S6 Table. Register sensitivity of the Greek archaizing measurement. The displacement of the Atticizers depends on which texts are treated as non-archaizing training material. Variant A uses the corpus manifest’s own labels; B additionally reclassifies authors independently described as classicizing in the literary-historical record; C discards register labels entirely and trains only on texts composed before 50 CE, prior to the Second Sophistic. Dates CE; a negative shift means the text is dated earlier than it was written.

Training set	Texts	Tested	MAE	ρ	Mean shift	Too early
A as-received manifest labels	42	14	145	+0.69	-206	11/14
B strict: classicizing authors reclassified	34	22	101	+0.80	-297	22/22
C date-blind: train only on $j \leq 50$ CE	26	11	120	+0.67	-397	11/11

S1 Appendix. Source partitions. Chapter-and-verse specification of every documentary source, sub-stratum and poem dated in this study (Table 8).

S1 Fig. Undated units under the reference specification. Point estimates with 68% and 90% jackknife+ intervals for all 19 undated units (Fig 6).

S2 Table. Power analysis. Intraclass correlation, design effect and effective sample size by passage size (Table 9).

S3 Table. Configuration search. Combinations of passage size, feature set and calibration with out-of-sample metrics, including the uncalibrated variants (Table 10).

Table 15. Sensitivity of each permutation p -value to the stopping point. Permutation runs are expensive, and a run stopped after its result is known invites the objection that it was stopped favourably. Both nulls are therefore reported at a stopping point declared in advance, and this table gives the p -value that each prefix of the same draw sequence would have produced. The within-genre p falls below 0.05 before 100 draws and stays there; the free-shuffle p is below 0.01 throughout. The exact value drifts, but neither conclusion depends on where the run was stopped.

Null	Draws	p
Within-genre shuffle, genre-controlled ρ	50	0.0588
	100	0.0396
	150	0.0331
	200	0.0249
	250	0.0199
	300	0.0166
Free shuffle, raw ρ	100	0.0099
	200	0.0050
	300	0.0033

S4 Table. Feature leverage. Years of apparent date per standard-deviation shift for every feature, by family (Tables 11 and 12). 1061 1062

S5 Table. Greek corpus. All texts with dates, registers, genres and token counts (Table 13). 1063

S6 Table. Register sensitivity of the Greek archaizing measurement. Displacement of the Atticizers under three definitions of the non-archaizing training set (Table 14). 1064 1065

S7 Table. Permutation p -values by stopping point. Each null's p -value computed on every prefix of its draw sequence, so that the dependence of the conclusion on where the run stopped can be inspected directly (Table 15). 1066 1067 1068

S8 Table. Undated units, numerical. The same estimates in tabular form, with passage counts and the probability of post-exilic composition (Table 19). 1069 1070

S9 Table. Every analytic specification. All 48 combinations of passage size, feature screen, weighting and calibration, with per-source estimates and the validation statistic for each (Table 16). 1071 1072

S11 Table. Block separations under the genre screen. Each within-source block comparison recomputed at three levels of the genre-diagnostic feature screen, showing that the separations do not survive it (Table 18). 1073 1074 1075

S10 Table. Ablations designed to break the result. Refits with the lexical or morphosyntactic feature family removed, and with the six anchors lacking an external synchronism dropped (Table 17). 1076 1077

Data Availability 1078

All code, extracted feature matrices, model outputs, and the scripts that regenerate every table, figure and numerical claim in this article are available at <https://github.com/adairaar/diachronic-hebrew>. The repository includes `run_pipeline.py`, which executes the analysis end to end and verifies its own output (see *Computational reproducibility* below). 1079 1080 1081 1082

Two inputs are third-party and are not redistributed. The BHS database is available under its own license from the Eep Talstra Centre for Bible and Computer and is retrieved automatically by the Text-Fabric access layer on first run; the repository records the version used. Greek texts derive from the 1083 1084 1085

Table 16. Every analytic specification. All 48 combinations of passage size, feature screen (share of features retained after dropping the most genre-diagnostic), inverse-density weighting and variance-matched calibration. Each row is a complete leave-one-book-out run. ρ is raw Spearman correlation with true date, $\rho|g$ the same with genre held fixed, MAE the mean absolute error in years, and the last three columns the point estimates for the sources in BCE. Rows marked $\dagger$ fall below the $\rho|g > 0.20$ validation threshold and are excluded from every figure quoted in the text. Sorted by validation statistic.

Words	Keep	Wt	Cal	ρ	ρg	MAE	JE	D	P
700	50%	yes	var	+0.80	+0.66	89	561	467	450
700	50%	yes	none	+0.80	+0.66	94	537	477	464
700	50%	no	var	+0.77	+0.59	100	570	480	476
700	50%	no	none	+0.77	+0.59	99	558	498	496
500	50%	yes	var	+0.77	+0.58	95	615	450	414
500	50%	yes	none	+0.77	+0.58	91	553	468	445
500	50%	no	var	+0.74	+0.57	100	599	433	420
500	50%	no	none	+0.74	+0.57	96	560	477	475
1000	50%	yes	var	+0.76	+0.43	90	583	451	498
1000	50%	yes	none	+0.76	+0.43	90	585	483	522
400	50%	yes	none	+0.68	+0.43	103	512	464	478
400	50%	yes	var	+0.68	+0.43	102	502	421	442
500	75%	yes	var	+0.73	+0.41	100	480	370	324
500	75%	yes	none	+0.73	+0.41	90	500	426	397
500	100%	yes	var	+0.66	+0.40	118	425	378	344
500	100%	yes	none	+0.66	+0.40	94	464	438	419
400	75%	no	none	+0.68	+0.40	87	470	440	420
400	75%	no	var	+0.68	+0.40	89	410	364	332
1000	50%	no	none	+0.75	+0.40	95	599	500	546
1000	50%	no	var	+0.75	+0.40	92	581	446	504
400	75%	yes	none	+0.67	+0.39	86	450	422	394
400	75%	yes	var	+0.67	+0.39	97	400	353	308
400	50%	no	var	+0.61	+0.38	111	495	419	455
400	50%	no	none	+0.61	+0.38	110	529	483	502
1000	75%	yes	var	+0.71	+0.38	85	471	408	358
1000	75%	yes	none	+0.71	+0.38	78	493	444	408
700	75%	yes	none	+0.66	+0.37	91	471	424	394
700	75%	yes	var	+0.66	+0.37	99	452	382	335
1000	75%	no	none	+0.71	+0.37	83	512	464	422
1000	75%	no	var	+0.71	+0.37	87	472	410	350
700	75%	no	var	+0.66	+0.36	101	456	374	324
700	75%	no	none	+0.66	+0.36	94	492	443	412
700	100%	yes	var	+0.63	+0.33	119	418	376	335
700	100%	yes	none	+0.63	+0.33	93	456	435	411
500	75%	no	none	+0.69	+0.33	96	519	445	422
500	75%	no	var	+0.69	+0.33	104	483	363	330
500	100%	no	none	+0.65	+0.30	99	480	452	444
500	100%	no	var	+0.65	+0.30	119	409	353	335
400	100%	no	var	+0.63	+0.24	117	402	377	346
400	100%	no	none	+0.63	+0.24	102	477	466	450
700	100%	no	var	+0.58	+0.22	120	397	345	329
700	100%	no	none	+0.58	+0.22	100	471	447	438
1000	100%	no	var	+0.61	+0.20	124	392	369	287
1000	100%	no	none	+0.61	+0.20	107	473	458	410
1000 $\dagger$	100%	yes	var	+0.61	+0.16	115	436	411	334
1000 $\dagger$	100%	yes	none	+0.61	+0.16	99	463	444	386
400 $\dagger$	100%	yes	var	+0.58	+0.13	116	405	393	341
400 $\dagger$	100%	yes	none	+0.58	+0.13	100	459	452	426

open First1KGreek and Perseus collections and are retrieved by a script in the repository. No other data are required.

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Table 17. Ablations designed to break the result. Each variant removes something the result might depend on and refits from scratch. Comparing B against C isolates which feature family carries the signal; D and E remove the anchors whose dates rest on literary rather than external grounds. Note that D also strips four late prophetic books and so restricts the within-genre chronological range, which cannot be separated from the circularity it is meant to test.

Variant	Features	Books	MAE	ρ	$\rho _{\text{genre}}$	ρ proph
Baseline: all features, all anchors	578	25	118	+0.665	+0.404	+0.378
Morphosyntax only (lexemes removed)	328	25	130	+0.538	+0.140	+0.059
Lexemes only (morphosyntax removed)	250	25	104	+0.678	+0.458	+0.429
Externally anchored books only	577	19	111	+0.592	+0.052	+0.102
Morphosyntax only, external anchors only	327	19	115	+0.618	+0.109	+0.212

Table 18. Block separations under the genre screen. Difference in median passage estimate between the conventionally distinguished blocks of each source, recomputed with the most genre-diagnostic features removed. A positive gap places the first-named block earlier. Screening improves the model’s genre-controlled ordering of the anchors (third column) while removing the separations entirely, and reversing the Deuteronomic one. Widening intervals reflect the smaller feature set; the sign change in Deuteronomy does not.

Feature set	p feats	$\rho _{\text{genre}}$	Comparison	Gap	95% CI	p
All features	578	+0.404	Deuteronomic law code vs frame	+97	+27 to +129	0.007
			Holiness Code vs Leviticus 1-16	+79	+38 to +145	0.016
			Genesis JE vs Exodus JE	+71	+13 to +120	0.039
Drop top 25%	434	+0.422	Deuteronomic law code vs frame	+7	-78 to +73	0.584
			Holiness Code vs Leviticus 1-16	-53	-214 to +52	0.150
			Genesis JE vs Exodus JE	+10	-96 to +115	0.995
Drop top 50%	289	+0.511	Deuteronomic law code vs frame	-156	-253 to -32	0.005
			Holiness Code vs Leviticus 1-16	-17	-222 to +86	0.254
			Genesis JE vs Exodus JE	+26	-106 to +157	0.432

Competing Interests

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Table 19. Undated units. Dates BCE; intervals are 68% conformal, calibrated on leave-one-book-out residuals. No unit here contributed to training, feature selection, weighting, regularization or calibration. The three poems are extracted at verse precision (see text).

Unit	Words	Pass.	Est.	68% interval	P(post-ex.)
<i>Archaic poems</i>					
Song of Deborah	448	1	818	1016–585	0.08
Song of the Sea	228	1	782	1021–560	0.12
Song of Moses	576	1	602	796–375	0.50
<i>Documentary sources</i>					
JE composite	37,750	74	425	619–173	0.86
<i>Sub-strata</i>					
Genesis JE	22,407	44	436	631–193	0.84
Exodus JE	11,780	23	363	554–138	0.96
Numbers JE	3,563	7	444	642–180	0.84
D source	20,128	39	378	571–113	0.94
D law code	7,353	14	421	619–158	0.84
D frame	11,993	23	332	530–101	0.96
P source	54,435	106	344	535–72	0.96
Leviticus P	10,496	21	319	520–60	0.96
Holiness Code	5,958	12	393	589–129	0.92
Jeremiah Dtr prose	15,197	30	445	646–190	0.84
<i>Pentateuch as received</i>					
Genesis	28,762	56	420	621–169	0.88
Exodus	23,748	47	357	551–108	0.96
Leviticus	17,099	33	338	536–67	0.96
Numbers	23,188	45	360	558–85	0.96
Deuteronomy	20,128	39	378	571–113	0.94

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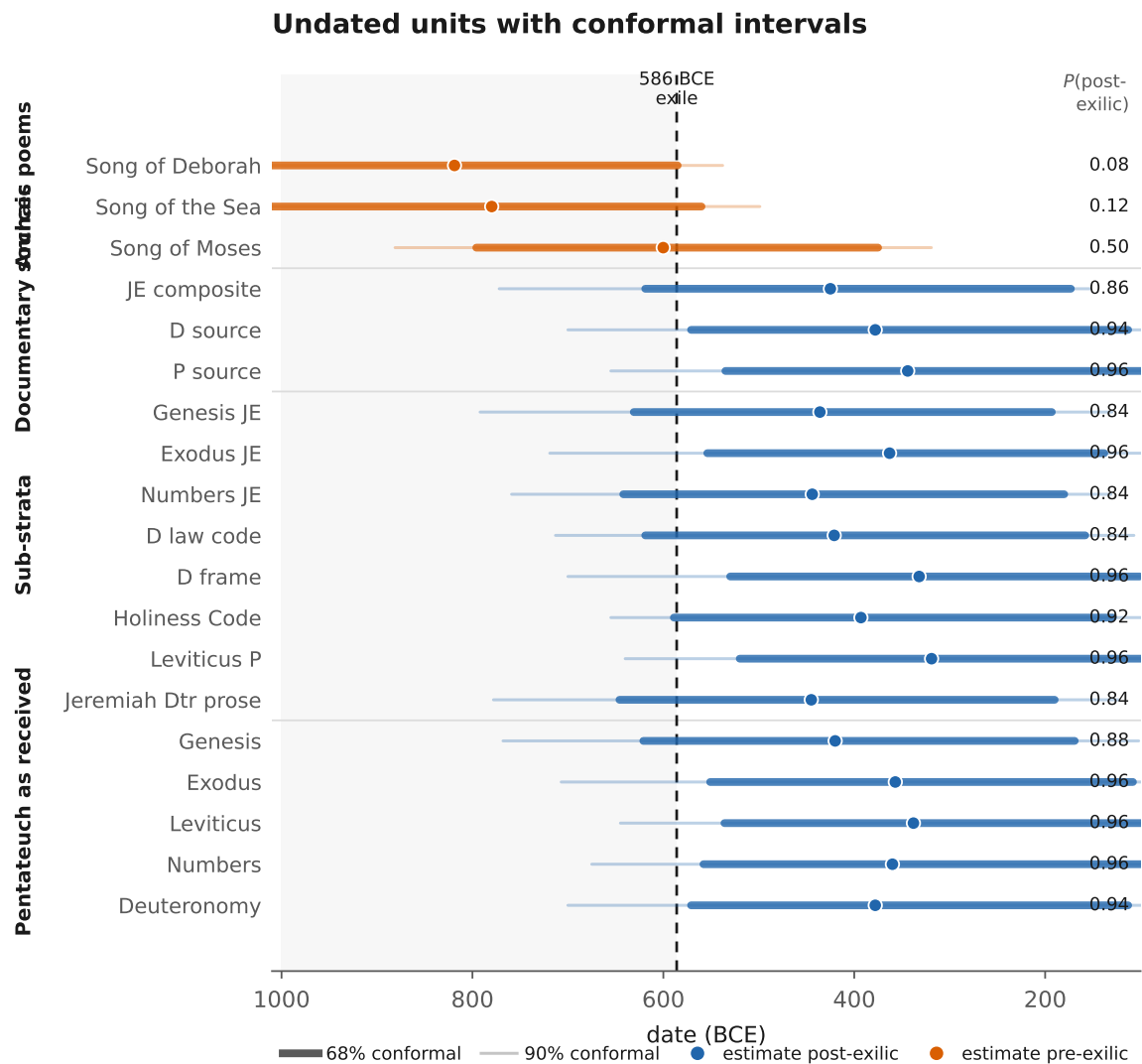


Fig 6. S1 Fig. Undated units under the reference specification. Thick bars are 68% jackknife+ intervals, thin bars 90%; the dashed line is the exile of 586 BCE. The three poems are extracted at verse precision. No unit shown contributed to training, feature selection, weighting, regularization or calibration. These are the estimates from one configuration among the 48 reported in Fig 3, and carry that configuration's arbitrariness.

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